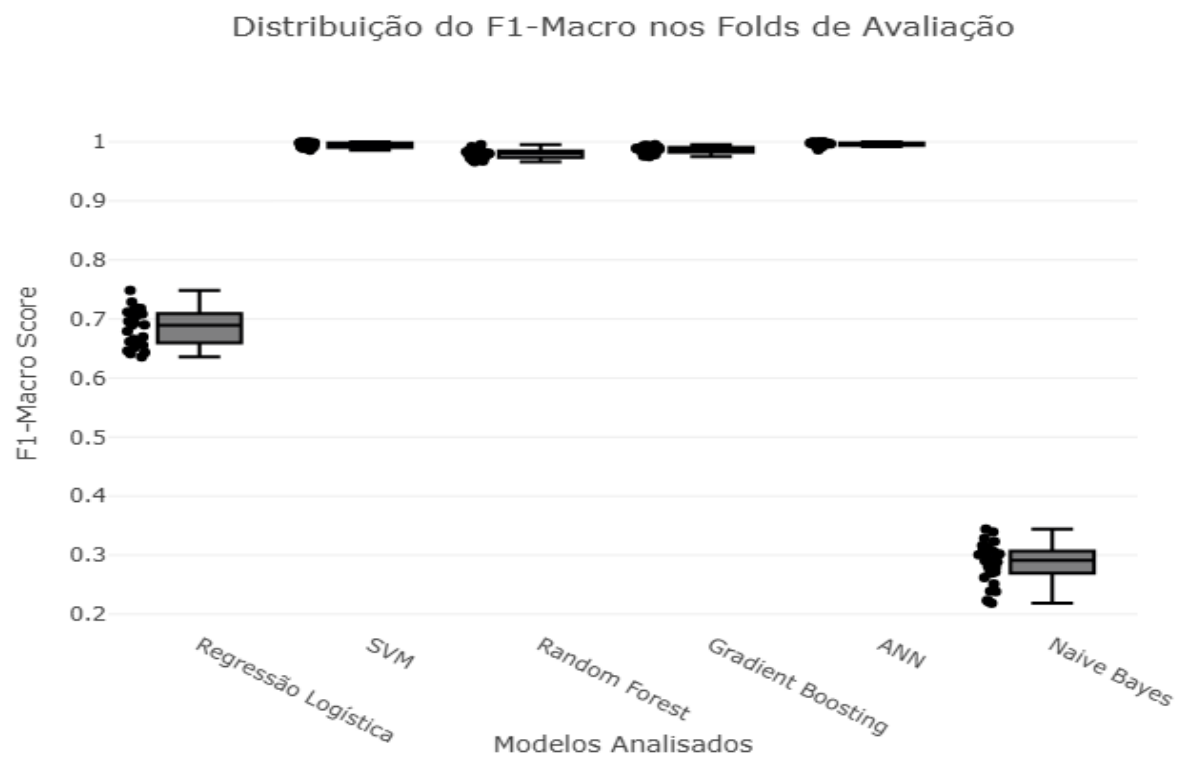


Relatório Técnico de Modelagem e Auditoria (Classificação)

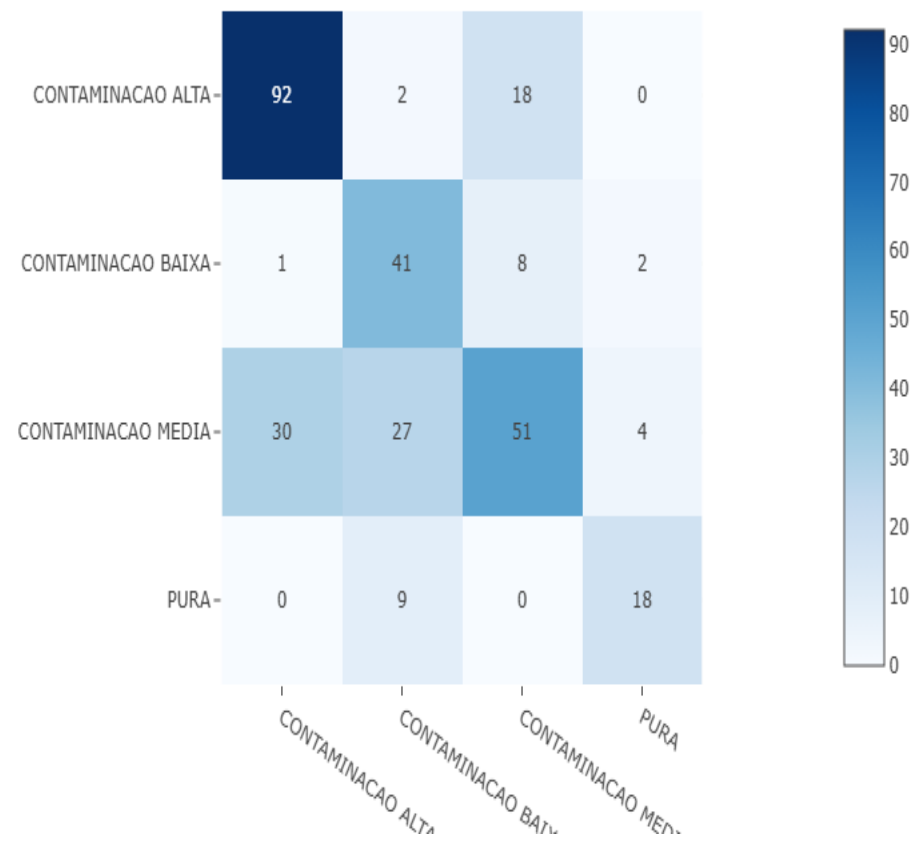
1. Desempenho Global (Nested CV / Teste Cego)

Modelo Utilizado	Acc (CV)	BAcc (CV)	F1-Macro (CV)	Acc (Externo)	BAcc (Externo)	F1 (Externo)	Configuração
Modelo A	0.9972908355261296	0.9956935425685426	0.996600114661576	0.7953795379537953	0.8475529100529101	0.8178706541134813	{'activation': 'tanh', 'alpha': 0.1, 'hidden_layer_sizes': (100, 100), 'solver': 'lbfgs'}
Modelo B	0.9902279093455565	0.9885536916786918	0.9894627220358939	0.7920792079207921	0.8382936507936508	0.8161329442983563	{'learning_rate': 0.1, 'max_depth': 10, 'n_estimators': 200, 'subsampling': 0.5}
Modelo C	0.9875045977987155	0.986603157824088	0.9875319752042037	0.768976897689769	0.8296957671957672	0.799804664832478	{'class_weight': None, 'max_depth': None, 'min_samples_split': 2, 'n_estimators': 100}
Modelo D	0.9956603853662678	0.99428360990861	0.9951525223969374	0.7524752475247525	0.8185350529100529	0.7854360948207323	{'C': 100, 'class_weight': None, 'gamma': 1.0, 'kernel': 'rbf'}
Modelo E	0.6974613077554254	0.687626297582693	0.6895310079649821	0.6666666666666666	0.6829784798534798	0.6636244536253135	{'C': 100.0, 'class_weight': None, 'solver': 'lbfgs'}
Modelo F	0.2992105933282404	0.3017293295636319	0.2776223017578444	0.25412541254125415	0.26742470492470494	0.22612169819736053	Padrão

Análise Estatística: Estabilidade (Teste de Friedman)



Diagnóstico Analítico: Regressão Logística



Auditoria de Falsos Positivos/Negativos (101 falhas)

tulo	TeorDensimetro	NivelAcucar	Temperatura	CPJ_10k	CPJ_4k7	CPJ_3k19	CPJ_1k	CPJ_824R	Real (Gabarito)	Classe Prevista	Probabilidade Estimada (%)	Probab: CONTAMINACAO ALTA (%)	Probab: CONTAMINACAO BAIXA (%)	Probab: CONTAMINACAO MEDIA (%)	Probab: PURA (%)
	38	5	19.1	21221.1	23551.2	23752.2	25341.0	25454.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	43.25	41.65	12.68	43.25	2.42
	38	5	19.1	20878.2	23656.1	23867.3	25486.0	25601.2	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	39.36	33.87	16.54	39.36	10.24
	38	5	19.1	20851.0	23662.1	23869.0	25492.2	25601.9	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	40.27	30.77	17.29	40.27	11.67
	38	5	19.2	20991.2	23628.0	23838.5	25411.1	25530.6	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	36.88	35.12	18.08	36.88	9.92
	38	5	20.9	21030.4	23860.9	24074.8	25951.3	26087.1	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	44.77	42.39	12.41	44.77	0.42

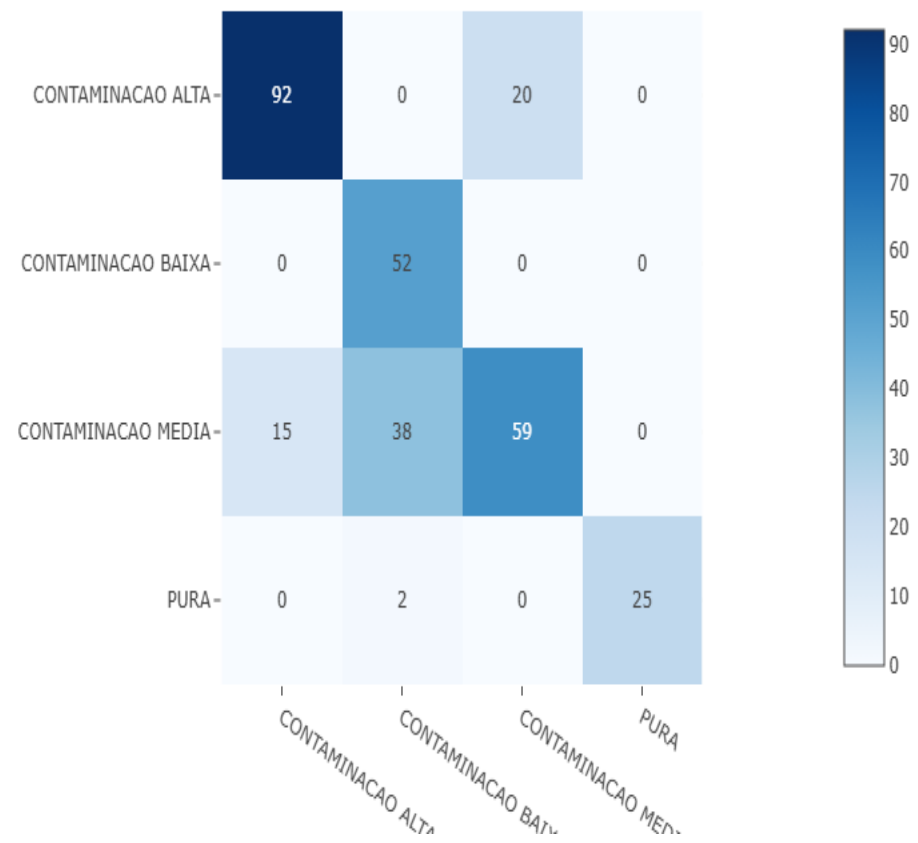
	38	5	19.2	21547.7	23675.6	23886.2	25448.6	25558.4	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	43.35	43.35	19.81	35.01	1.84
	38	5	19.2	21626.6	23670.6	23882.2	25411.1	25518.9	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	44.35	44.35	20.98	32.64	2.03
	38	5	19.2	20477.3	23795.9	24010.0	25613.6	25732.7	CONTAMINACAO MEDIA	PURA	56.94	9.15	18.12	15.79	56.94
	38	5	19.2	21816.7	23633.1	23846.0	25355.1	25464.9	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	51.07	51.07	17.26	30.72	0.96
	38	5	19.2	20483.8	23811.6	24024.8	25655.1	25773.8	CONTAMINACAO MEDIA	PURA	50.04	10.46	20.48	19.02	50.04
	38	5	20.8	21334.4	23872.3	24086.3	25863.9	25986.6	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	43.87	43.87	17.42	38.15	0.56
	38	5	26.0	20758.5	24244.4	24483.1	26394.1	26575.0	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	46.29	46.29	31.11	20.52	2.09
	38	5	26.0	21069.9	24141.3	24372.5	26358.2	26522.7	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	66.26	66.26	10.36	23.25	0.13
	38	5	26.0	20729.5	24255.4	24498.1	26441.0	26586.4	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	60.49	60.49	16.85	20.74	1.92
	38	5	26.0	20766.1	24237.4	24477.0	26401.0	26560.5	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	55.51	55.51	21.52	21.13	1.85
	38	5	26.0	20655.0	24261.9	24491.4	26425.5	26585.7	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	41.52	41.52	31.85	23.45	3.18
	38	5	26.0	20783.0	24233.4	24476.5	26467.1	26637.6	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	61.64	61.64	15.5	22.24	0.63
	38	5	26.0	20823.6	24168.5	24413.3	26373.1	26537.8	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	69.31	69.31	10.23	19.95	0.52
	38	5	26.0	20797.7	24214.9	24452.7	26430.3	26600.7	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	58.57	58.57	17.31	23.45	0.68
	38	5	26.0	20779.0	24226.7	24468.1	26448.8	26616.2	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	60.95	60.95	15.88	22.44	0.73
	38	5	26.0	20587.6	24275.5	24512.5	26414.1	26597.9	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	40.86	34.84	40.86	19.12	5.18
	38	5	26.0	20805.8	24171.0	24412.9	26366.3	26531.9	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	66.15	66.15	12.16	21.04	0.65
	38	5	26.2	20894.0	24182.3	24418.5	26394.6	26561.2	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	64.32	64.32	12.91	22.4	0.37
	38	5	26.2	20976.2	24166.2	24403.1	26387.9	26554.6	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	67.89	67.89	10.38	21.53	0.21
	38	5	26.2	20848.7	24181.1	24420.3	26382.2	26547.3	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	65.65	65.65	12.54	21.3	0.52
	38	5	26.2	21033.0	24147.7	24380.7	26363.2	26528.1	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	67.61	67.61	10.05	22.18	0.16
	38	5	26.2	20916.1	24175.0	24412.9	26392.5	26557.3	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	67.17	67.17	11.0	21.53	0.3
	38	5	26.2	20993.4	24159.1	24393.6	26377.4	26545.8	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	66.57	66.57	10.97	22.26	0.19
	38	5	26.2	21013.4	24152.1	24389.1	26370.7	26537.5	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	69.47	69.47	9.4	20.96	0.17
	38	5	26.2	20870.1	24175.2	24415.7	26387.0	26554.6	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	67.48	67.48	11.23	20.91	0.39
	38	5	26.2	20935.3	24169.3	24406.7	26383.9	26551.3	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	66.93	66.93	11.21	21.59	0.28
	38	5	19.2	21639.4	23708.1	23921.9	25461.4	25586.9	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	38.72	38.72	28.11	31.34	1.83
	38	5	19.2	20888.0	23848.7	24062.3	25678.3	25794.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	34.3	16.95	34.3	25.89	22.86
	38	5	19.2	21178.0	23805.3	24017.9	25622.0	25736.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	33.52	26.06	33.52	32.01	8.41
	38	5	19.2	22343.4	23634.1	23842.3	25314.5	25421.7	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	53.68	53.68	18.78	27.35	0.2

	38	5	20.7	21606.3	23925.0	24140.3	25919.3	26038.8	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	44.0	44.0	20.18	35.56	0.26
	38	5	20.7	22583.3	23808.9	24021.7	25744.2	25867.2	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	63.74	63.74	9.76	26.5	0.0
	38	5	19.1	22388.8	23712.2	23925.6	25403.7	25517.5	CONTAMINACAO BAIXA	CONTAMINACAO ALTA	47.74	47.74	26.76	25.28	0.22
	38	5	18.8	20481.0	24179.1	24397.8	26208.2	26339.3	PURA	CONTAMINACAO BAIXA	53.2	3.94	53.2	14.83	28.02
	38	5	18.8	20863.6	24086.3	24308.7	26012.8	26145.0	PURA	CONTAMINACAO BAIXA	58.05	7.07	58.05	16.32	18.55
	38	5	18.8	21003.4	24077.7	24297.6	25989.2	26107.2	PURA	CONTAMINACAO BAIXA	57.82	8.77	57.82	18.4	15.01
	38	5	19.0	20815.3	24077.0	24298.0	25999.6	26119.9	PURA	CONTAMINACAO BAIXA	51.02	7.89	51.02	17.07	24.03
	41	0	25.3	18913.1	25143.2	25389.7	27718.3	27912.8	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	38.64	32.62	23.07	38.64	5.67
	41	0	25.5	18973.3	25147.0	25389.6	27719.9	27933.1	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	38.06	28.21	29.78	38.06	3.95
	41	0	17.6	20131.7	24970.7	25227.8	27320.4	27481.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	39.44	20.54	39.44	38.86	1.16
	41	0	18.0	21033.6	24797.0	25055.3	27094.9	27264.2	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	47.83	47.83	15.84	36.31	0.02
	41	0	17.6	18831.8	25206.4	25447.8	27764.1	27948.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	56.11	2.35	56.11	25.02	16.52
	41	0	24.3	17443.1	25384.3	25561.7	28337.0	28647.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	71.27	0.35	71.27	24.46	3.93
	41	0	24.3	17507.4	25388.9	25570.7	28312.4	28604.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	68.56	0.46	68.56	25.12	5.87
	41	0	24.3	17356.9	25404.0	25580.9	28343.3	28661.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	75.45	0.23	75.45	18.8	5.52
	41	0	24.3	17484.2	25394.9	25574.4	28341.0	28638.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	69.27	0.42	69.27	25.71	4.6
	41	0	24.3	17638.0	25380.0	25548.4	28251.2	28570.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	82.29	0.17	82.29	14.55	2.98
	41	0	24.3	17516.4	25385.2	25563.7	28302.7	28618.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	75.67	0.3	75.67	19.99	4.04
	41	0	24.3	17470.3	25387.6	25565.0	28334.8	28636.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	70.02	0.38	70.02	25.35	4.25
	41	0	24.3	17406.0	25377.6	25554.2	28341.0	28651.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	69.4	0.37	69.4	26.28	3.95
	41	0	24.3	17556.2	25378.8	25555.9	28283.0	28570.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	69.55	0.42	69.55	24.38	5.64
	41	0	24.3	17425.4	25382.3	25559.7	28341.7	28651.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	70.21	0.36	70.21	25.5	3.93
	41	0	25.2	17518.9	25448.7	25628.3	28348.0	28620.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	69.23	0.44	69.23	20.6	9.72
	41	0	25.2	17479.0	25439.5	25615.0	28334.3	28613.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	70.93	0.35	70.93	19.09	9.63
	41	0	25.2	17447.5	25441.0	25617.4	28344.1	28627.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	70.93	0.35	70.93	19.09	9.63
	41	0	25.2	17405.7	25430.7	25608.1	28365.7	28660.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	70.8	0.38	70.8	21.67	7.15
	41	0	25.2	17550.3	25447.2	25626.7	28318.7	28577.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	66.34	0.47	66.34	19.85	13.34
	41	0	25.2	17421.9	25440.6	25618.1	28346.5	28634.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	71.11	0.34	71.11	18.68	9.87
	41	0	25.2	17536.0	25445.4	25623.9	28327.9	28594.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	68.5	0.43	68.5	19.88	11.19
	41	0	25.2	17580.9	25459.6	25645.5	28339.2	28619.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	72.04	0.44	72.04	17.62	9.89

	41	0	25.2	17525.5	25457.9	25633.8	28263.6	28628.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	90.43	0.05	90.43	4.5	5.01
	41	0	25.2	17498.2	25447.2	25627.2	28337.7	28629.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	74.03	0.33	74.03	16.83	8.81
	41	0	17.4	20490.7	24967.7	25219.4	27370.4	27550.3	CONTAMINACAO BAIXA	CONTAMINACAO MEDIA	41.82	19.89	38.18	41.82	0.12
	41	0	17.5	20358.4	24941.6	25195.4	27282.4	27447.8	CONTAMINACAO BAIXA	CONTAMINACAO MEDIA	39.72	21.78	38.06	39.72	0.44
	41	0	19.9	20161.0	25022.2	25269.2	27409.1	27568.3	CONTAMINACAO BAIXA	CONTAMINACAO MEDIA	38.54	25.37	35.47	38.54	0.61
	41	0	19.8	18409.3	25359.8	25595.6	27944.8	28114.9	CONTAMINACAO BAIXA	PURA	51.97	0.91	36.93	10.2	51.97
	39	0	19.1	17281.4	26207.0	26384.7	29150.9	29405.9	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	44.29	35.89	18.41	44.29	1.41
	39	0	19.1	17548.1	26107.9	26274.6	29101.2	29374.4	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	50.58	38.07	11.23	50.58	0.11
	39	0	25.9	15845.8	25860.5	25928.5	28662.7	29036.8	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	39.03	3.92	38.81	39.03	18.23
	39	0	26.0	17355.9	25906.9	26027.0	29062.6	29462.8	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	53.52	42.06	4.42	53.52	0.0
	39	0	26.0	17283.7	25952.5	26082.7	29154.7	29553.7	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	50.1	46.26	3.64	50.1	0.0
	39	0	26.0	17408.6	25889.7	26003.5	29025.1	29428.2	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	56.07	38.71	5.21	56.07	0.0
	39	0	26.0	17326.8	25919.2	26047.3	29099.1	29503.4	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	50.03	46.28	3.69	50.03	0.0
	39	0	26.0	17291.1	25953.8	26086.3	29160.5	29557.8	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	48.48	48.23	3.29	48.48	0.0
	39	0	26.0	17303.8	25936.5	26061.8	29121.0	29522.1	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	52.36	43.48	4.16	52.36	0.0
	39	0	26.0	17288.9	25943.9	26072.4	29138.5	29538.4	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	50.81	45.41	3.78	50.81	0.0
	39	0	18.8	18341.6	25611.4	25757.7	28196.7	28406.2	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	67.42	67.42	2.27	30.25	0.06
	39	0	18.8	18411.4	25598.1	25742.2	28175.5	28389.4	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	66.92	66.92	2.38	30.66	0.04
	39	0	15.4	16572.4	26106.8	26200.6	28837.3	29085.9	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	50.55	0.38	50.55	14.47	34.6
	39	0	15.9	17243.4	25995.6	26110.0	28709.1	28945.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	47.69	4.15	47.69	41.59	6.57
	39	0	19.1	16794.0	26079.1	26204.6	28963.8	29262.5	CONTAMINACAO BAIXA	CONTAMINACAO MEDIA	47.03	6.9	42.24	47.03	3.83
	39	0	19.1	16931.1	26051.0	26176.6	28906.0	29200.2	CONTAMINACAO BAIXA	CONTAMINACAO MEDIA	48.08	8.24	40.61	48.08	3.07
	39	0	15.5	16446.8	26266.0	26360.2	29000.2	29239.6	CONTAMINACAO BAIXA	PURA	53.8	0.1	41.06	5.04	53.8
	39	0	15.5	17521.3	26048.5	26179.6	28829.1	29075.5	CONTAMINACAO BAIXA	CONTAMINACAO MEDIA	46.58	6.68	45.11	46.58	1.63
	39	0	15.4	16470.1	26301.4	26392.4	28975.0	29317.3	PURA	CONTAMINACAO BAIXA	75.27	0.01	75.27	1.42	23.3
	39	0	18.9	16327.4	26201.3	26303.0	29061.7	29383.1	PURA	CONTAMINACAO BAIXA	67.44	0.63	67.44	16.16	15.77
	39	0	26.2	15810.8	25663.1	25674.9	28076.0	28556.7	PURA	CONTAMINACAO BAIXA	64.59	0.05	64.59	2.85	32.5
	34	15	19.1	16663.5	26163.3	26299.1	29315.2	29690.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	53.21	36.36	10.4	53.21	0.03
	34	15	21.5	16360.9	26214.3	26263.6	28945.6	29238.3	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	64.39	2.87	64.39	21.57	11.17
	34	15	21.5	16421.1	26190.2	26235.9	28859.9	29129.8	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	58.9	2.9	58.9	19.72	18.48

	34	15	21.5	16044.4	26041.3	26090.7	28631.1	28880.0	CONTAMINACAO MEDIA	PURA	67.93	2.28	18.28	11.5	67.93
	34	15	21.5	16199.9	25997.7	26046.9	28613.4	28852.1	CONTAMINACAO MEDIA	PURA	45.28	6.28	22.84	25.6	45.28
	34	15	19.1	16981.7	25822.1	25841.2	28383.0	28649.9	CONTAMINACAO BAIXA	CONTAMINACAO MEDIA	47.24	5.02	45.76	47.24	1.99
	34	15	19.1	17383.0	25700.9	25737.1	28238.0	28495.6	CONTAMINACAO BAIXA	CONTAMINACAO MEDIA	60.15	17.01	22.37	60.15	0.47
	34	15	19.1	15948.8	26127.1	26129.3	28713.2	28997.2	PURA	CONTAMINACAO BAIXA	51.66	0.11	51.66	5.5	42.73
	34	15	19.1	16370.5	25991.4	25997.8	28555.8	28838.0	PURA	CONTAMINACAO BAIXA	67.3	0.59	67.3	16.47	15.64

Diagnóstico Analítico: SVM



Auditoria de Falsos Positivos/Negativos (75 falhas)

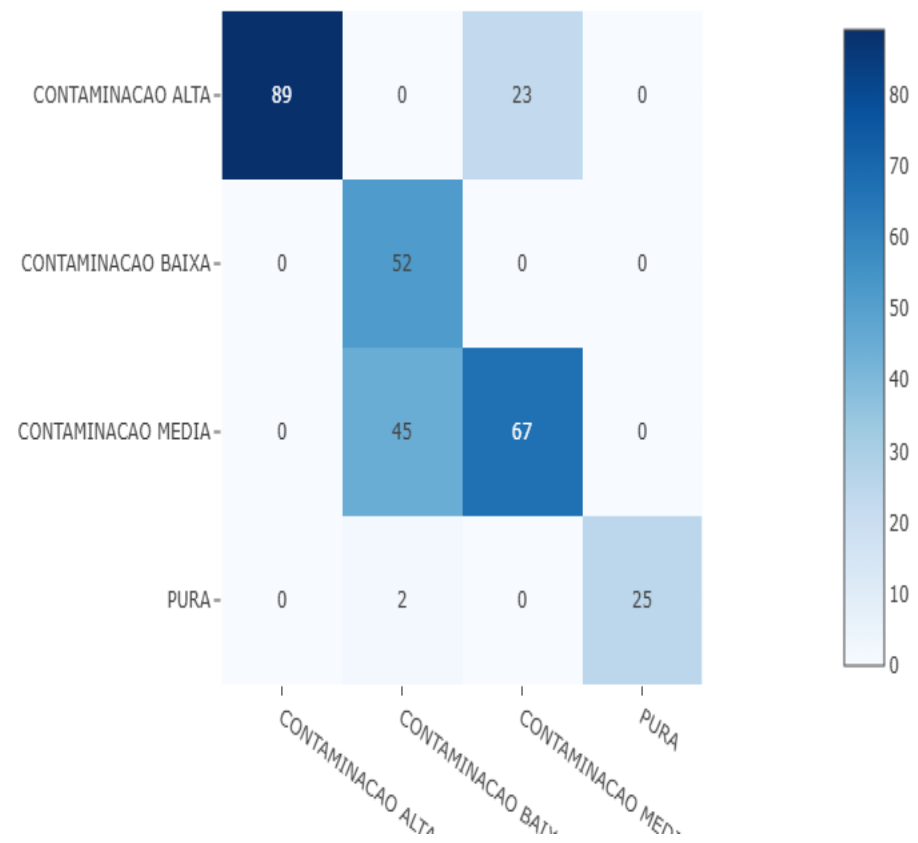
Titulo	TeorDensimetro	NivelAcucar	Temperatura	CPJ_10k	CPJ_4k7	CPJ_3k19	CPJ_1k	CPJ_824R	Real (Gabarito)	Classe Prevista	Probabilidade Estimada (%)	Probab: CONTAMINACAO ALTA (%)	Probab: CONTAMINACAO BAIXA (%)	Probab: CONTAMINACAO MEDIA (%)	Probab: PURA (%)
	38	5	26.0	20758.5	24244.4	24483.1	26394.1	26575.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.12	0.66	99.12	0.2	0.02
	38	5	26.0	21069.9	24141.3	24372.5	26358.2	26522.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	66.16	33.25	66.16	0.44	0.14
	38	5	26.0	20729.5	24255.4	24498.1	26441.0	26586.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.5	0.29	99.5	0.2	0.02
	38	5	26.0	20766.1	24237.4	24477.0	26401.0	26560.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	98.93	0.85	98.93	0.2	0.02
	38	5	26.0	20655.0	24261.9	24491.4	26425.5	26585.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.51	0.28	99.51	0.2	0.02

	38	5	26.0	20783.0	24233.4	24476.5	26467.1	26637.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.42	0.36	99.42	0.2	0.02
	38	5	26.0	20823.6	24168.5	24413.3	26373.1	26537.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	90.27	9.43	90.27	0.26	0.04
	38	5	26.0	20797.7	24214.9	24452.7	26430.3	26600.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	98.66	1.12	98.66	0.2	0.02
	38	5	26.0	20779.0	24226.7	24468.1	26448.8	26616.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.21	0.58	99.21	0.2	0.02
	38	5	26.0	20587.6	24275.5	24512.5	26414.1	26597.9	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.64	0.15	99.64	0.2	0.02
	38	5	26.0	20805.8	24171.0	24412.9	26366.3	26531.9	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	89.95	9.75	89.95	0.26	0.04
	38	5	26.2	20894.0	24182.3	24418.5	26394.6	26561.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	81.04	18.57	81.04	0.31	0.07
	38	5	26.2	20976.2	24166.2	24403.1	26387.9	26554.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	68.72	30.75	68.72	0.41	0.12
	38	5	26.2	20848.7	24181.1	24420.3	26382.2	26547.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	79.04	20.56	79.04	0.32	0.08
	38	5	26.2	21033.0	24147.7	24380.7	26363.2	26528.1	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	54.83	54.83	44.35	0.62	0.21
	38	5	26.2	20916.1	24175.0	24412.9	26392.5	26557.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	76.62	22.95	76.62	0.34	0.09
	38	5	26.2	20993.4	24159.1	24393.6	26377.4	26545.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	59.46	39.91	59.46	0.47	0.15
	38	5	26.2	21013.4	24152.1	24389.1	26370.7	26537.5	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	52.28	47.01	52.28	0.53	0.18
	38	5	26.2	20870.1	24175.2	24415.7	26387.0	26554.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	77.3	22.29	77.3	0.33	0.08
	38	5	26.2	20935.3	24169.3	24406.7	26383.9	26551.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	70.67	28.84	70.67	0.39	0.11
	41	0	25.3	18910.7	25141.9	25397.6	27731.1	27903.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	79.8	0.29	19.61	79.8	0.29
	41	0	25.3	18920.3	25146.3	25401.3	27739.9	27895.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	79.99	0.29	19.43	79.99	0.3
	41	0	25.3	18913.1	25143.2	25389.7	27718.3	27912.8	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	79.88	0.29	19.53	79.88	0.29
	41	0	25.3	18911.7	25144.4	25403.3	27738.0	27893.7	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	79.84	0.29	19.57	79.84	0.3
	41	0	25.3	18934.6	25148.1	25401.0	27740.2	27897.4	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	80.22	0.28	19.2	80.22	0.3
	41	0	25.3	18917.1	25139.7	25394.3	27725.2	27898.9	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	80.76	0.3	18.64	80.76	0.3
	41	0	25.3	18931.5	25138.6	25393.2	27722.5	27893.9	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	81.77	0.3	17.63	81.77	0.3
	41	0	25.3	18924.7	25136.2	25392.3	27721.3	27893.1	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	81.8	0.3	17.6	81.8	0.3
	41	0	25.3	18937.0	25134.0	25389.1	27713.4	27886.4	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	83.09	0.3	16.3	83.09	0.3
	41	0	25.3	18910.2	25144.2	25400.9	27735.2	27885.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	80.59	0.3	18.81	80.59	0.3
	41	0	25.5	18947.8	25149.8	25401.3	27748.3	27910.5	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	98.23	0.33	1.08	98.23	0.35
	41	0	25.5	18952.7	25148.5	25405.3	27758.4	27920.3	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	98.19	0.31	1.15	98.19	0.35
	41	0	25.5	18973.3	25147.0	25389.6	27719.9	27933.1	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	98.31	0.33	1.01	98.31	0.35
	41	0	25.5	19012.2	25153.9	25407.4	27779.2	27950.1	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	98.16	0.24	1.25	98.16	0.35
	41	0	25.5	18990.8	25156.1	25409.1	27779.6	27949.5	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	98.09	0.25	1.31	98.09	0.35

	41	0	25.5	18960.0	25154.4	25408.2	27767.3	27927.3	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	98.13	0.29	1.22	98.13	0.35
	41	0	25.5	18974.7	25155.1	25409.4	27774.4	27942.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	98.09	0.27	1.29	98.09	0.35
	41	0	25.5	19063.6	25152.1	25404.9	27773.7	27949.4	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	98.32	0.23	1.11	98.32	0.34
	41	0	25.5	18940.4	25149.4	25404.0	27749.6	27908.2	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	98.2	0.34	1.1	98.2	0.35
	41	0	25.5	19034.3	25154.0	25406.9	27778.9	27953.7	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	98.2	0.23	1.22	98.2	0.35
	41	0	24.3	17443.1	25384.3	25561.7	28337.0	28647.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	88.51	8.94	88.51	1.92	0.63
	41	0	24.3	17507.4	25388.9	25570.7	28312.4	28604.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	90.69	7.2	90.69	1.6	0.5
	41	0	24.3	17356.9	25404.0	25580.9	28343.3	28661.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	85.22	11.62	85.22	2.34	0.82
	41	0	24.3	17484.2	25394.9	25574.4	28341.0	28638.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	89.0	8.54	89.0	1.85	0.6
	41	0	24.3	17638.0	25380.0	25548.4	28251.2	28570.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	94.61	4.16	94.61	0.94	0.29
	41	0	24.3	17516.4	25385.2	25563.7	28302.7	28618.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	91.05	6.92	91.05	1.55	0.48
	41	0	24.3	17470.3	25387.6	25565.0	28334.8	28636.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	89.19	8.39	89.19	1.83	0.59
	41	0	24.3	17406.0	25377.6	25554.2	28341.0	28651.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	87.82	9.51	87.82	2.01	0.67
	41	0	24.3	17556.2	25378.8	25555.9	28283.0	28570.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	92.89	5.48	92.89	1.25	0.38
	41	0	24.3	17425.4	25382.3	25559.7	28341.7	28651.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	88.03	9.33	88.03	1.98	0.66
	41	0	25.2	17518.9	25448.7	25628.3	28348.0	28620.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	86.58	8.27	86.58	4.58	0.57
	41	0	25.2	17479.0	25439.5	25615.0	28334.3	28613.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	86.6	8.23	86.6	4.61	0.56
	41	0	25.2	17447.5	25441.0	25617.4	28344.1	28627.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	85.63	9.11	85.63	4.63	0.62
	41	0	25.2	17405.7	25430.7	25608.1	28365.7	28660.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	84.52	10.22	84.52	4.55	0.7
	41	0	25.2	17550.3	25447.2	25626.7	28318.7	28577.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	87.89	6.96	87.89	4.67	0.48
	41	0	25.2	17421.9	25440.6	25618.1	28346.5	28634.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	85.0	9.68	85.0	4.66	0.66
	41	0	25.2	17536.0	25445.4	25623.9	28327.9	28594.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	87.52	7.35	87.52	4.62	0.51
	41	0	25.2	17580.9	25459.6	25645.5	28339.2	28619.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	87.19	7.72	87.19	4.56	0.53
	41	0	25.2	17525.5	25457.9	25633.8	28263.6	28628.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	87.4	7.45	87.4	4.63	0.51
	41	0	25.2	17498.2	25447.2	25627.2	28337.7	28629.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	86.33	8.5	86.33	4.59	0.58
	39	0	22.5	17041.6	25409.1	25447.2	28013.6	28377.2	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	41.13	16.68	41.13	38.56	3.62
	39	0	22.5	16986.9	25413.4	25451.1	28015.4	28380.3	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	40.31	16.56	40.31	39.65	3.48
	39	0	22.5	17070.6	25411.0	25447.7	28014.4	28380.2	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	41.47	16.65	41.47	38.21	3.68
	39	0	22.5	16963.6	25409.6	25443.0	28008.1	28373.5	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	40.3	16.88	40.3	39.33	3.49
	39	0	22.5	16929.8	25420.1	25455.5	28025.5	28391.1	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	40.91	16.41	39.36	40.91	3.32

	39	0	22.5	16907.8	25416.5	25448.9	28012.7	28379.1	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	40.46	16.78	39.42	40.46	3.34
	39	0	22.5	16928.0	25412.0	25448.2	28012.6	28380.0	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	40.13	16.78	39.7	40.13	3.39
	39	0	22.5	17169.5	25405.8	25443.0	27973.3	28367.7	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	43.32	17.34	43.32	35.31	4.04
	39	0	22.5	17113.6	25407.1	25446.2	28008.4	28377.9	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	42.16	16.84	42.16	37.2	3.8
	39	0	22.5	16947.0	25409.4	25445.7	28010.7	28375.3	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	40.03	16.84	40.03	39.68	3.45
	39	0	23.4	16627.6	25335.8	25351.2	27849.2	28193.4	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	48.2	33.88	12.43	48.2	5.5
	39	0	23.4	16665.7	25347.4	25363.8	27868.3	28217.9	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	49.5	33.09	12.09	49.5	5.31
	39	0	23.4	16694.9	25357.7	25377.3	27890.2	28239.5	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	50.66	32.4	11.78	50.66	5.15
	39	0	15.4	16470.1	26301.4	26392.4	28975.0	29317.3	PURA	CONTAMINACAO BAIXA	99.36	0.13	99.36	0.0	0.51
	34	15	19.1	16370.5	25991.4	25997.8	28555.8	28838.0	PURA	CONTAMINACAO BAIXA	54.43	0.02	44.68	0.87	54.43

Diagnóstico Analítico: Random Forest



Auditoria de Falsos Positivos/Negativos (70 falhas)

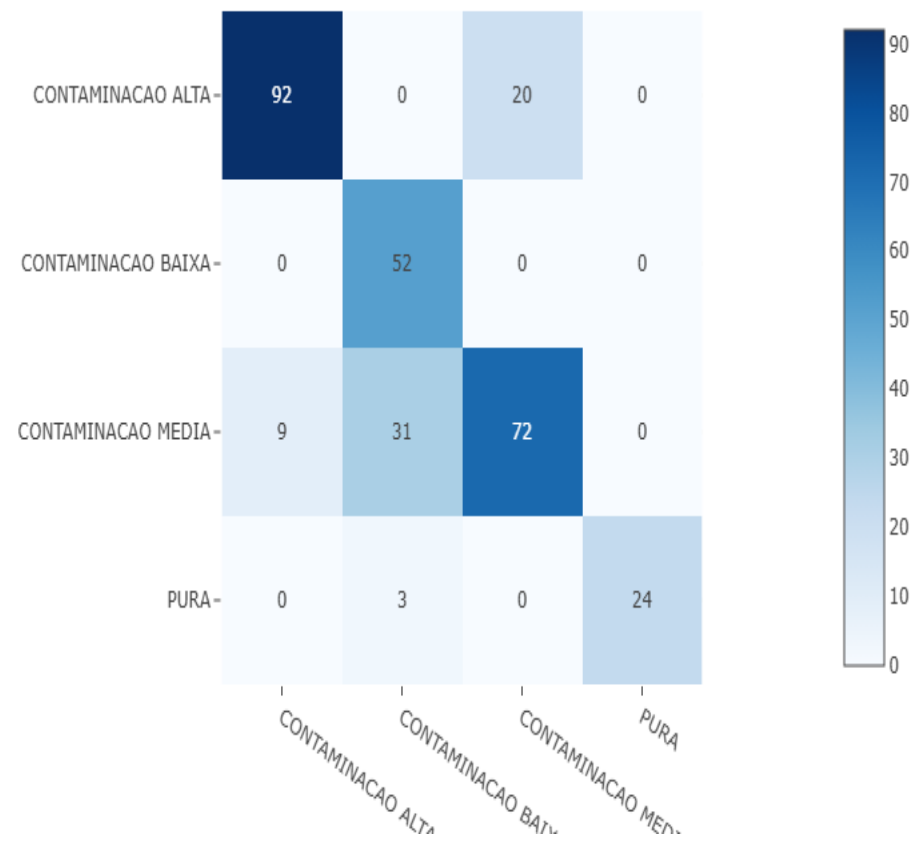
tulo	TeorDensimetro	NivelAcucar	Temperatura	CPJ_10k	CPJ_4k7	CPJ_3k19	CPJ_1k	CPJ_824R	Real (Gabarito)	Classe Prevista	Probabilidade Estimada (%)	Probab: CONTAMINACAO ALTA (%)	Probab: CONTAMINACAO BAIXA (%)	Probab: CONTAMINACAO MEDIA (%)	Probab: PURA (%)
	38	5	26.0	20758.5	24244.4	24483.1	26394.1	26575.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	90.0	0.0	90.0	0.0	10.0
	38	5	26.0	21069.9	24141.3	24372.5	26358.2	26522.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	78.0	14.0	78.0	0.0	8.0
	38	5	26.0	20729.5	24255.4	24498.1	26441.0	26586.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	82.0	2.0	82.0	0.0	16.0
	38	5	26.0	20766.1	24237.4	24477.0	26401.0	26560.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	88.0	0.0	88.0	0.0	12.0
	38	5	26.0	20655.0	24261.9	24491.4	26425.5	26585.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	92.0	0.0	92.0	0.0	8.0

	38	5	26.0	20783.0	24233.4	24476.5	26467.1	26637.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	90.0	0.0	90.0	0.0	10.0
	38	5	26.0	20823.6	24168.5	24413.3	26373.1	26537.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	86.0	6.0	86.0	0.0	8.0
	38	5	26.0	20797.7	24214.9	24452.7	26430.3	26600.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	92.0	0.0	92.0	0.0	8.0
	38	5	26.0	20779.0	24226.7	24468.1	26448.8	26616.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	90.0	0.0	90.0	0.0	10.0
	38	5	26.0	20587.6	24275.5	24512.5	26414.1	26597.9	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	68.0	0.0	68.0	0.0	32.0
	38	5	26.0	20805.8	24171.0	24412.9	26366.3	26531.9	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	86.0	6.0	86.0	0.0	8.0
	38	5	26.2	20894.0	24182.3	24418.5	26394.6	26561.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	70.0	20.0	70.0	0.0	10.0
	38	5	26.2	20976.2	24166.2	24403.1	26387.9	26554.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	72.0	20.0	72.0	0.0	8.0
	38	5	26.2	20848.7	24181.1	24420.3	26382.2	26547.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	68.0	24.0	68.0	0.0	8.0
	38	5	26.2	21033.0	24147.7	24380.7	26363.2	26528.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	60.0	34.0	60.0	0.0	6.0
	38	5	26.2	20916.1	24175.0	24412.9	26392.5	26557.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	70.0	20.0	70.0	0.0	10.0
	38	5	26.2	20993.4	24159.1	24393.6	26377.4	26545.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	68.0	24.0	68.0	0.0	8.0
	38	5	26.2	21013.4	24152.1	24389.1	26370.7	26537.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	64.0	30.0	64.0	0.0	6.0
	38	5	26.2	20870.1	24175.2	24415.7	26387.0	26554.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	72.0	20.0	72.0	0.0	8.0
	38	5	26.2	20935.3	24169.3	24406.7	26383.9	26551.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	66.0	24.0	66.0	0.0	10.0
	41	0	25.3	18910.7	25141.9	25397.6	27731.1	27903.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	68.0	0.0	30.0	68.0	2.0
	41	0	25.3	18920.3	25146.3	25401.3	27739.9	27895.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	62.0	0.0	34.0	62.0	4.0
	41	0	25.3	18913.1	25143.2	25389.7	27718.3	27912.8	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	66.0	0.0	30.0	66.0	4.0
	41	0	25.3	18911.7	25144.4	25403.3	27738.0	27893.7	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	62.0	0.0	34.0	62.0	4.0
	41	0	25.3	18934.6	25148.1	25401.0	27740.2	27897.4	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	62.0	0.0	34.0	62.0	4.0
	41	0	25.3	18917.1	25139.7	25394.3	27725.2	27898.9	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	68.0	0.0	30.0	68.0	2.0
	41	0	25.3	18931.5	25138.6	25393.2	27722.5	27893.9	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	66.0	0.0	32.0	66.0	2.0
	41	0	25.3	18924.7	25136.2	25392.3	27721.3	27893.1	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	66.0	0.0	32.0	66.0	2.0
	41	0	25.3	18937.0	25134.0	25389.1	27713.4	27886.4	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	66.0	0.0	32.0	66.0	2.0
	41	0	25.3	18910.2	25144.2	25400.9	27735.2	27885.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	62.0	0.0	34.0	62.0	4.0
	41	0	25.5	18947.8	25149.8	25401.3	27748.3	27910.5	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	84.0	0.0	14.0	84.0	2.0
	41	0	25.5	18952.7	25148.5	25405.3	27758.4	27920.3	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	82.0	0.0	12.0	82.0	6.0
	41	0	25.5	18973.3	25147.0	25389.6	27719.9	27933.1	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	86.0	0.0	10.0	86.0	4.0
	41	0	25.5	19012.2	25153.9	25407.4	27779.2	27950.1	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	84.0	0.0	12.0	84.0	4.0
	41	0	25.5	18990.8	25156.1	25409.1	27779.6	27949.5	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	86.0	0.0	12.0	86.0	2.0

	41	0	25.5	18960.0	25154.4	25408.2	27767.3	27927.3	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	84.0	0.0	12.0	84.0	4.0
	41	0	25.5	18974.7	25155.1	25409.4	27774.4	27942.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	86.0	0.0	12.0	86.0	2.0
	41	0	25.5	19063.6	25152.1	25404.9	27773.7	27949.4	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	78.0	0.0	20.0	78.0	2.0
	41	0	25.5	18940.4	25149.4	25404.0	27749.6	27908.2	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	84.0	0.0	14.0	84.0	2.0
	41	0	25.5	19034.3	25154.0	25406.9	27778.9	27953.7	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	78.0	0.0	20.0	78.0	2.0
	41	0	24.3	17443.1	25384.3	25561.7	28337.0	28647.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	48.0	4.0	48.0	38.0	10.0
	41	0	24.3	17507.4	25388.9	25570.7	28312.4	28604.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	48.0	4.0	48.0	38.0	10.0
	41	0	24.3	17356.9	25404.0	25580.9	28343.3	28661.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	44.0	4.0	44.0	42.0	10.0
	41	0	24.3	17484.2	25394.9	25574.4	28341.0	28638.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	48.0	4.0	48.0	38.0	10.0
	41	0	24.3	17638.0	25380.0	25548.4	28251.2	28570.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	50.0	4.0	50.0	34.0	12.0
	41	0	24.3	17516.4	25385.2	25563.7	28302.7	28618.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	50.0	4.0	50.0	36.0	10.0
	41	0	24.3	17470.3	25387.6	25565.0	28334.8	28636.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	48.0	4.0	48.0	38.0	10.0
	41	0	24.3	17406.0	25377.6	25554.2	28341.0	28651.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	48.0	4.0	48.0	38.0	10.0
	41	0	24.3	17556.2	25378.8	25555.9	28283.0	28570.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	50.0	4.0	50.0	34.0	12.0
	41	0	24.3	17425.4	25382.3	25559.7	28341.7	28651.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	48.0	4.0	48.0	38.0	10.0
	39	0	26.0	17355.9	25906.9	26027.0	29062.6	29462.8	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	46.0	34.0	20.0	46.0	0.0
	39	0	26.0	17408.6	25889.7	26003.5	29025.1	29428.2	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	46.0	34.0	20.0	46.0	0.0
	39	0	26.0	17326.8	25919.2	26047.3	29099.1	29503.4	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	50.0	30.0	20.0	50.0	0.0
	39	0	22.5	16963.6	25409.6	25443.0	28008.1	28373.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	48.0	2.0	48.0	48.0	2.0
	39	0	22.5	16929.8	25420.1	25455.5	28025.5	28391.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	48.0	2.0	48.0	48.0	2.0
	39	0	22.5	16907.8	25416.5	25448.9	28012.7	28379.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	48.0	2.0	48.0	48.0	2.0
	39	0	22.5	16928.0	25412.0	25448.2	28012.6	28380.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	48.0	2.0	48.0	48.0	2.0
	39	0	22.5	16947.0	25409.4	25445.7	28010.7	28375.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	48.0	2.0	48.0	48.0	2.0
	39	0	23.4	16755.8	25381.6	25408.3	27924.0	28277.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	58.0	2.0	58.0	38.0	2.0
	39	0	23.4	16837.8	25404.4	25437.6	27987.3	28352.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	56.0	2.0	56.0	40.0	2.0
	39	0	23.4	16627.6	25335.8	25351.2	27849.2	28193.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	46.0	2.0	46.0	46.0	6.0
	39	0	23.4	16732.7	25368.3	25389.3	27906.2	28256.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	58.0	2.0	58.0	38.0	2.0
	39	0	23.4	16665.7	25347.4	25363.8	27868.3	28217.9	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	52.0	2.0	52.0	42.0	4.0
	39	0	23.4	16814.2	25389.1	25417.2	27970.8	28333.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	56.0	2.0	56.0	40.0	2.0
	39	0	23.4	16694.9	25357.7	25377.3	27890.2	28239.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	54.0	2.0	54.0	42.0	2.0

39	0	23.4	16860.2	25407.4	25436.6	27991.0	28354.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	56.0	2.0	56.0	40.0	2.0
39	0	23.4	16764.8	25376.7	25400.9	27919.7	28271.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	58.0	2.0	58.0	38.0	2.0
39	0	23.4	16766.4	25394.8	25424.2	27926.0	28297.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	56.0	2.0	56.0	40.0	2.0
39	0	15.4	16470.1	26301.4	26392.4	28975.0	29317.3	PURA	CONTAMINACAO BAIXA	62.0	2.0	62.0	0.0	36.0
34	15	19.1	16370.5	25991.4	25997.8	28555.8	28838.0	PURA	CONTAMINACAO BAIXA	92.0	0.0	92.0	8.0	0.0

Diagnóstico Analítico: Gradient Boosting



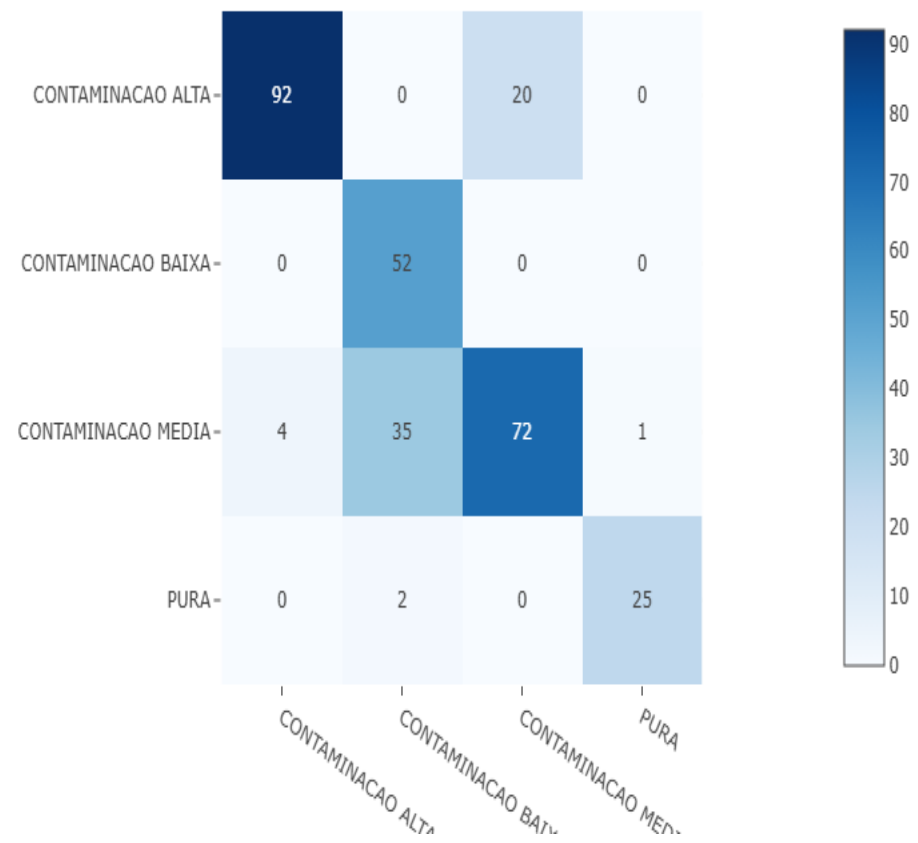
Auditoria de Falsos Positivos/Negativos (63 falhas)

Titulo	TeorDensimetro	NivelAcucar	Temperatura	CPJ_10k	CPJ_4k7	CPJ_3k19	CPJ_1k	CPJ_824R	Real (Gabarito)	Classe Prevista	Probabilidade Estimada (%)	Probab: CONTAMINACAO ALTA (%)	Probab: CONTAMINACAO BAIXA (%)	Probab: CONTAMINACAO MEDIA (%)	Probab: PURA (%)
	38	5	26.0	20758.5	24244.4	24483.1	26394.1	26575.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	38	5	26.0	21069.9	24141.3	24372.5	26358.2	26522.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.33	0.67	99.33	0.0	0.0
	38	5	26.0	20729.5	24255.4	24498.1	26441.0	26586.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	38	5	26.0	20766.1	24237.4	24477.0	26401.0	26560.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	38	5	26.0	20655.0	24261.9	24491.4	26425.5	26585.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0

	38	5	26.0	20783.0	24233.4	24476.5	26467.1	26637.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	38	5	26.0	20823.6	24168.5	24413.3	26373.1	26537.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.99	0.01	99.99	0.0	0.0
	38	5	26.0	20797.7	24214.9	24452.7	26430.3	26600.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	38	5	26.0	20779.0	24226.7	24468.1	26448.8	26616.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	38	5	26.0	20587.6	24275.5	24512.5	26414.1	26597.9	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.87	0.13	99.87	0.0	0.0
	38	5	26.0	20805.8	24171.0	24412.9	26366.3	26531.9	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.97	0.03	99.97	0.0	0.0
	38	5	26.2	20894.0	24182.3	24418.5	26394.6	26561.2	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	99.99	99.99	0.01	0.0	0.0
	38	5	26.2	20976.2	24166.2	24403.1	26387.9	26554.6	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	99.99	99.99	0.01	0.0	0.0
	38	5	26.2	20848.7	24181.1	24420.3	26382.2	26547.3	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	99.99	99.99	0.01	0.0	0.0
	38	5	26.2	21033.0	24147.7	24380.7	26363.2	26528.1	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	100.0	100.0	0.0	0.0	0.0
	38	5	26.2	20916.1	24175.0	24412.9	26392.5	26557.3	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	99.99	99.99	0.01	0.0	0.0
	38	5	26.2	20993.4	24159.1	24393.6	26377.4	26545.8	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	100.0	100.0	0.0	0.0	0.0
	38	5	26.2	21013.4	24152.1	24389.1	26370.7	26537.5	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	100.0	100.0	0.0	0.0	0.0
	38	5	26.2	20870.1	24175.2	24415.7	26387.0	26554.6	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	99.99	99.99	0.01	0.0	0.0
	38	5	26.2	20935.3	24169.3	24406.7	26383.9	26551.3	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	100.0	100.0	0.0	0.0	0.0
	41	0	25.3	18910.7	25141.9	25397.6	27731.1	27903.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.81	0.0	0.19	99.81	0.0
	41	0	25.3	18920.3	25146.3	25401.3	27739.9	27895.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.81	0.0	0.19	99.81	0.0
	41	0	25.3	18913.1	25143.2	25389.7	27718.3	27912.8	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.78	0.0	0.22	99.78	0.0
	41	0	25.3	18911.7	25144.4	25403.3	27738.0	27893.7	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.81	0.0	0.19	99.81	0.0
	41	0	25.3	18934.6	25148.1	25401.0	27740.2	27897.4	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.81	0.0	0.19	99.81	0.0
	41	0	25.3	18917.1	25139.7	25394.3	27725.2	27898.9	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.81	0.0	0.19	99.81	0.0
	41	0	25.3	18931.5	25138.6	25393.2	27722.5	27893.9	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.81	0.0	0.19	99.81	0.0
	41	0	25.3	18924.7	25136.2	25392.3	27721.3	27893.1	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.78	0.0	0.22	99.78	0.0
	41	0	25.3	18937.0	25134.0	25389.1	27713.4	27886.4	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.78	0.0	0.22	99.78	0.0
	41	0	25.3	18910.2	25144.2	25400.9	27735.2	27885.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.81	0.0	0.19	99.81	0.0
	41	0	25.5	18947.8	25149.8	25401.3	27748.3	27910.5	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	100.0	0.0	0.0	100.0	0.0
	41	0	25.5	18952.7	25148.5	25405.3	27758.4	27920.3	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	100.0	0.0	0.0	100.0	0.0
	41	0	25.5	18973.3	25147.0	25389.6	27719.9	27933.1	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	100.0	0.0	0.0	100.0	0.0
	41	0	25.5	19012.2	25153.9	25407.4	27779.2	27950.1	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	100.0	0.0	0.0	100.0	0.0
	41	0	25.5	18990.8	25156.1	25409.1	27779.6	27949.5	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	100.0	0.0	0.0	100.0	0.0

	41	0	25.5	18960.0	25154.4	25408.2	27767.3	27927.3	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	100.0	0.0	0.0	100.0	0.0
	41	0	25.5	18974.7	25155.1	25409.4	27774.4	27942.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	100.0	0.0	0.0	100.0	0.0
	41	0	25.5	19063.6	25152.1	25404.9	27773.7	27949.4	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	100.0	0.0	0.0	100.0	0.0
	41	0	25.5	18940.4	25149.4	25404.0	27749.6	27908.2	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	100.0	0.0	0.0	100.0	0.0
	41	0	25.5	19034.3	25154.0	25406.9	27778.9	27953.7	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	100.0	0.0	0.0	100.0	0.0
	41	0	24.3	17443.1	25384.3	25561.7	28337.0	28647.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.98	0.0	99.98	0.02	0.0
	41	0	24.3	17507.4	25388.9	25570.7	28312.4	28604.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.98	0.0	99.98	0.02	0.0
	41	0	24.3	17356.9	25404.0	25580.9	28343.3	28661.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.92	0.0	99.92	0.08	0.0
	41	0	24.3	17484.2	25394.9	25574.4	28341.0	28638.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.98	0.0	99.98	0.02	0.0
	41	0	24.3	17638.0	25380.0	25548.4	28251.2	28570.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.99	0.0	99.99	0.01	0.0
	41	0	24.3	17516.4	25385.2	25563.7	28302.7	28618.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.98	0.0	99.98	0.02	0.0
	41	0	24.3	17470.3	25387.6	25565.0	28334.8	28636.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.98	0.0	99.98	0.02	0.0
	41	0	24.3	17406.0	25377.6	25554.2	28341.0	28651.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.96	0.0	99.96	0.04	0.0
	41	0	24.3	17556.2	25378.8	25555.9	28283.0	28570.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.99	0.0	99.99	0.01	0.0
	41	0	24.3	17425.4	25382.3	25559.7	28341.7	28651.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.97	0.0	99.97	0.03	0.0
	39	0	23.4	16755.8	25381.6	25408.3	27924.0	28277.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	98.1	0.0	98.1	1.9	0.0
	39	0	23.4	16837.8	25404.4	25437.6	27987.3	28352.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	98.11	0.0	98.11	1.89	0.0
	39	0	23.4	16627.6	25335.8	25351.2	27849.2	28193.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	94.54	0.0	94.54	5.45	0.0
	39	0	23.4	16732.7	25368.3	25389.3	27906.2	28256.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	96.57	0.0	96.57	3.43	0.0
	39	0	23.4	16665.7	25347.4	25363.8	27868.3	28217.9	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	95.73	0.0	95.73	4.27	0.01
	39	0	23.4	16814.2	25389.1	25417.2	27970.8	28333.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	98.86	0.0	98.86	1.14	0.0
	39	0	23.4	16694.9	25357.7	25377.3	27890.2	28239.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	96.36	0.0	96.36	3.63	0.0
	39	0	23.4	16860.2	25407.4	25436.6	27991.0	28354.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	97.81	0.0	97.81	2.19	0.0
	39	0	23.4	16764.8	25376.7	25400.9	27919.7	28271.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	97.68	0.0	97.68	2.32	0.0
	39	0	23.4	16766.4	25394.8	25424.2	27926.0	28297.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	98.82	0.0	98.82	1.18	0.0
	39	0	15.4	16470.1	26301.4	26392.4	28975.0	29317.3	PURA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	39	0	15.5	16753.3	26185.2	26272.1	28767.1	28966.1	PURA	CONTAMINACAO BAIXA	89.74	0.0	89.74	0.0	10.25
	34	15	19.1	16370.5	25991.4	25997.8	28555.8	28838.0	PURA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0

Diagnóstico Analítico: ANN



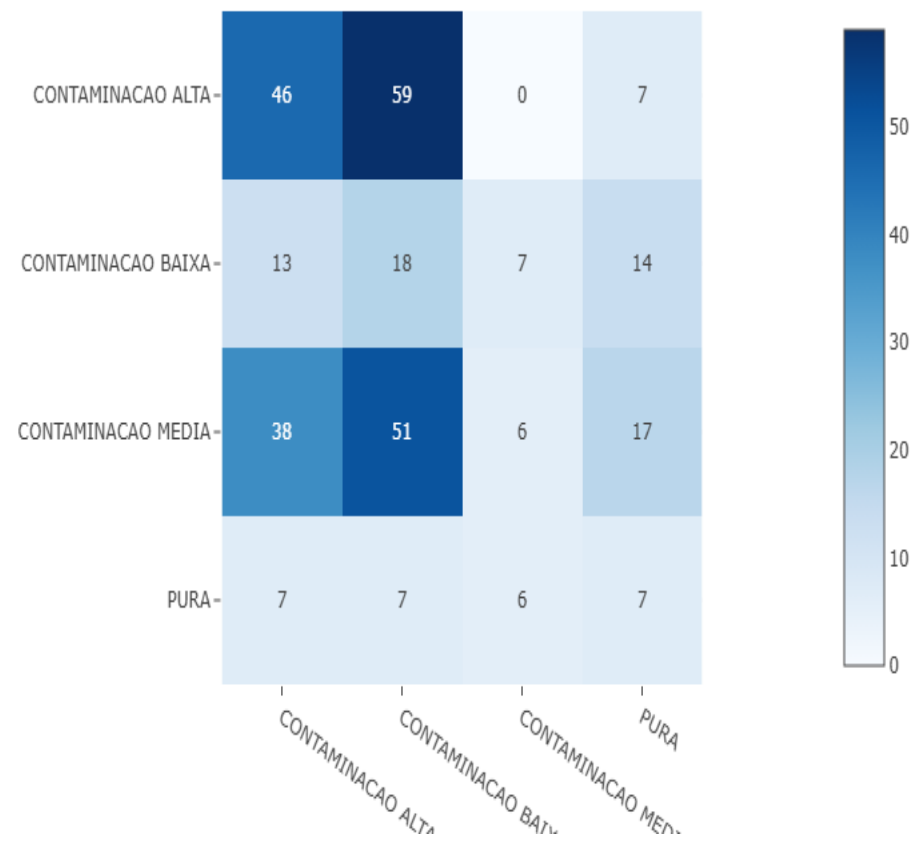
Auditoria de Falsos Positivos/Negativos (62 falhas)

Titulo	TeorDensimetro	NivelAcucar	Temperatura	CPJ_10k	CPJ_4k7	CPJ_3k19	CPJ_1k	CPJ_824R	Real (Gabarito)	Classe Prevista	Probabilidade Estimada (%)	Probab: CONTAMINACAO ALTA (%)	Probab: CONTAMINACAO BAIXA (%)	Probab: CONTAMINACAO MEDIA (%)	Probab: PURA (%)
	38	5	26.0	20758.5	24244.4	24483.1	26394.1	26575.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.97	0.03	99.97	0.0	0.0
	38	5	26.0	21069.9	24141.3	24372.5	26358.2	26522.7	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	80.07	80.07	19.93	0.0	0.0
	38	5	26.0	20729.5	24255.4	24498.1	26441.0	26586.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.99	0.01	99.99	0.0	0.0
	38	5	26.0	20766.1	24237.4	24477.0	26401.0	26560.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.95	0.05	99.95	0.0	0.0
	38	5	26.0	20655.0	24261.9	24491.4	26425.5	26585.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.99	0.01	99.99	0.0	0.0

	38	5	26.0	20783.0	24233.4	24476.5	26467.1	26637.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.93	0.07	99.93	0.0	0.0
	38	5	26.0	20823.6	24168.5	24413.3	26373.1	26537.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	86.1	13.9	86.1	0.0	0.0
	38	5	26.0	20797.7	24214.9	24452.7	26430.3	26600.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.63	0.37	99.63	0.0	0.0
	38	5	26.0	20779.0	24226.7	24468.1	26448.8	26616.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.88	0.12	99.88	0.0	0.0
	38	5	26.0	20587.6	24275.5	24512.5	26414.1	26597.9	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	38	5	26.0	20805.8	24171.0	24412.9	26366.3	26531.9	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	87.3	12.7	87.3	0.0	0.0
	38	5	26.2	20894.0	24182.3	24418.5	26394.6	26561.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	86.4	13.6	86.4	0.0	0.0
	38	5	26.2	20976.2	24166.2	24403.1	26387.9	26554.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	59.67	40.33	59.67	0.0	0.0
	38	5	26.2	20848.7	24181.1	24420.3	26382.2	26547.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	86.89	13.11	86.89	0.0	0.0
	38	5	26.2	21033.0	24147.7	24380.7	26363.2	26528.1	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	81.21	81.21	18.79	0.0	0.0
	38	5	26.2	20916.1	24175.0	24412.9	26392.5	26557.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	78.05	21.95	78.05	0.0	0.0
	38	5	26.2	20993.4	24159.1	24393.6	26377.4	26545.8	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	58.57	58.57	41.43	0.0	0.0
	38	5	26.2	21013.4	24152.1	24389.1	26370.7	26537.5	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	70.92	70.92	29.08	0.0	0.0
	38	5	26.2	20870.1	24175.2	24415.7	26387.0	26554.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	80.85	19.15	80.85	0.0	0.0
	38	5	26.2	20935.3	24169.3	24406.7	26383.9	26551.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	67.24	32.76	67.24	0.0	0.0
	41	0	25.3	18910.7	25141.9	25397.6	27731.1	27903.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	97.66	0.02	1.71	97.66	0.62
	41	0	25.3	18920.3	25146.3	25401.3	27739.9	27895.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	97.9	0.02	1.4	97.9	0.68
	41	0	25.3	18913.1	25143.2	25389.7	27718.3	27912.8	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	97.21	0.02	2.22	97.21	0.55
	41	0	25.3	18911.7	25144.4	25403.3	27738.0	27893.7	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	97.79	0.02	1.43	97.79	0.76
	41	0	25.3	18934.6	25148.1	25401.0	27740.2	27897.4	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	97.96	0.02	1.44	97.96	0.59
	41	0	25.3	18917.1	25139.7	25394.3	27725.2	27898.9	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	97.77	0.02	1.64	97.77	0.57
	41	0	25.3	18931.5	25138.6	25393.2	27722.5	27893.9	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	97.97	0.02	1.5	97.97	0.5
	41	0	25.3	18924.7	25136.2	25392.3	27721.3	27893.1	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	97.98	0.02	1.49	97.98	0.51
	41	0	25.3	18937.0	25134.0	25389.1	27713.4	27886.4	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	98.12	0.02	1.39	98.12	0.46
	41	0	25.3	18910.2	25144.2	25400.9	27735.2	27885.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	97.86	0.02	1.26	97.86	0.86
	41	0	25.5	18947.8	25149.8	25401.3	27748.3	27910.5	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.73	0.01	0.05	99.73	0.21
	41	0	25.5	18952.7	25148.5	25405.3	27758.4	27920.3	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.77	0.01	0.06	99.77	0.16
	41	0	25.5	18973.3	25147.0	25389.6	27719.9	27933.1	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.78	0.01	0.09	99.78	0.13
	41	0	25.5	19012.2	25153.9	25407.4	27779.2	27950.1	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.86	0.0	0.08	99.86	0.06
	41	0	25.5	18990.8	25156.1	25409.1	27779.6	27949.5	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.84	0.0	0.08	99.84	0.08

	41	0	25.5	18960.0	25154.4	25408.2	27767.3	27927.3	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.78	0.0	0.06	99.78	0.16
	41	0	25.5	18974.7	25155.1	25409.4	27774.4	27942.0	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.82	0.0	0.07	99.82	0.11
	41	0	25.5	19063.6	25152.1	25404.9	27773.7	27949.4	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.88	0.0	0.08	99.88	0.03
	41	0	25.5	18940.4	25149.4	25404.0	27749.6	27908.2	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.7	0.01	0.05	99.7	0.24
	41	0	25.5	19034.3	25154.0	25406.9	27778.9	27953.7	CONTAMINACAO ALTA	CONTAMINACAO MEDIA	99.87	0.0	0.08	99.87	0.04
	41	0	24.3	17443.1	25384.3	25561.7	28337.0	28647.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	41	0	24.3	17507.4	25388.9	25570.7	28312.4	28604.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	41	0	24.3	17356.9	25404.0	25580.9	28343.3	28661.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	41	0	24.3	17484.2	25394.9	25574.4	28341.0	28638.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	41	0	24.3	17638.0	25380.0	25548.4	28251.2	28570.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	41	0	24.3	17516.4	25385.2	25563.7	28302.7	28618.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	41	0	24.3	17470.3	25387.6	25565.0	28334.8	28636.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	41	0	24.3	17406.0	25377.6	25554.2	28341.0	28651.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	41	0	24.3	17556.2	25378.8	25555.9	28283.0	28570.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	41	0	24.3	17425.4	25382.3	25559.7	28341.7	28651.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	100.0	0.0	100.0	0.0	0.0
	41	0	25.2	17518.9	25448.7	25628.3	28348.0	28620.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	82.46	0.0	82.46	0.67	16.88
	41	0	25.2	17479.0	25439.5	25615.0	28334.3	28613.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	77.08	0.0	77.08	0.63	22.29
	41	0	25.2	17447.5	25441.0	25617.4	28344.1	28627.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	85.41	0.0	85.41	0.4	14.19
	41	0	25.2	17405.7	25430.7	25608.1	28365.7	28660.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	99.34	0.0	99.34	0.06	0.59
	41	0	25.2	17550.3	25447.2	25626.7	28318.7	28577.4	CONTAMINACAO MEDIA	PURA	71.73	0.0	27.08	1.18	71.73
	41	0	25.2	17421.9	25440.6	25618.1	28346.5	28634.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	88.05	0.0	88.05	0.3	11.65
	41	0	25.2	17536.0	25445.4	25623.9	28327.9	28594.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	52.65	0.0	52.65	1.14	46.21
	41	0	25.2	17580.9	25459.6	25645.5	28339.2	28619.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	77.44	0.0	77.44	0.87	21.69
	41	0	25.2	17525.5	25457.9	25633.8	28263.6	28628.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	59.63	0.0	59.63	0.4	39.98
	41	0	25.2	17498.2	25447.2	25627.2	28337.7	28629.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	86.42	0.0	86.42	0.45	13.13
	39	0	15.4	16470.1	26301.4	26392.4	28975.0	29317.3	PURA	CONTAMINACAO BAIXA	99.99	0.0	99.99	0.0	0.01
	34	15	19.1	16370.5	25991.4	25997.8	28555.8	28838.0	PURA	CONTAMINACAO BAIXA	67.58	0.0	67.58	0.04	32.38

Diagnóstico Analítico: Naive Bayes



Auditoria de Falsos Positivos/Negativos (226 falhas)

Titulo	TeorDensimetro	NivelAcucar	Temperatura	CPJ_10k	CPJ_4k7	CPJ_3k19	CPJ_1k	CPJ_824R	Real (Gabarito)	Classe Prevista	Probabilidade Estimada (%)	Probab: CONTAMINACAO ALTA (%)	Probab: CONTAMINACAO BAIXA (%)	Probab: CONTAMINACAO MEDIA (%)	Probab: PURA (%)
	38	5	19.2	21547.7	23675.6	23886.2	25448.6	25558.4	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	78.86	78.86	3.58	17.37	0.19
	38	5	19.2	21626.6	23670.6	23882.2	25411.1	25518.9	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	79.41	79.41	3.43	16.99	0.17
	38	5	19.2	20477.3	23795.9	24010.0	25613.6	25732.7	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	72.99	72.99	5.28	21.26	0.48
	38	5	19.2	21816.7	23633.1	23846.0	25355.1	25464.9	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	80.65	80.65	3.1	16.12	0.14
	38	5	19.2	20483.8	23811.6	24024.8	25655.1	25773.8	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	72.26	72.26	5.51	21.72	0.51

	38	5	20.8	21334.4	23872.3	24086.3	25863.9	25986.6	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	72.56	72.56	6.03	21.03	0.38
	38	5	20.8	20908.4	23941.4	24159.5	25965.5	26092.3	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	69.27	69.27	7.21	22.94	0.58
	38	5	26.0	20758.5	24244.4	24483.1	26394.1	26575.0	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	55.29	55.29	34.68	9.62	0.41
	38	5	26.0	21069.9	24141.3	24372.5	26358.2	26522.7	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	59.53	59.53	30.88	9.28	0.3
	38	5	26.0	20729.5	24255.4	24498.1	26441.0	26586.4	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	54.44	54.44	35.45	9.68	0.43
	38	5	26.0	20766.1	24237.4	24477.0	26401.0	26560.5	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	55.52	55.52	34.47	9.61	0.4
	38	5	26.0	20655.0	24261.9	24491.4	26425.5	26585.7	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	54.28	54.28	35.59	9.7	0.44
	38	5	26.0	20783.0	24233.4	24476.5	26467.1	26637.6	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	54.61	54.61	35.3	9.67	0.42
	38	5	26.0	20823.6	24168.5	24413.3	26373.1	26537.8	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	57.66	57.66	32.53	9.45	0.36
	38	5	26.0	20797.7	24214.9	24452.7	26430.3	26600.7	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	55.69	55.69	34.31	9.59	0.4
	38	5	26.0	20779.0	24226.7	24468.1	26448.8	26616.2	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	55.06	55.06	34.89	9.64	0.41
	38	5	26.0	20587.6	24275.5	24512.5	26414.1	26597.9	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	53.58	53.58	36.21	9.75	0.46
	38	5	26.0	20805.8	24171.0	24412.9	26366.3	26531.9	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	57.66	57.66	32.53	9.45	0.36
	38	5	26.2	20894.0	24182.3	24418.5	26394.6	26561.2	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	56.18	56.18	34.78	8.71	0.33
	38	5	26.2	20976.2	24166.2	24403.1	26387.9	26554.6	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	56.94	56.94	34.09	8.66	0.31
	38	5	26.2	20848.7	24181.1	24420.3	26382.2	26547.3	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	56.2	56.2	34.76	8.71	0.33
	38	5	26.2	21033.0	24147.7	24380.7	26363.2	26528.1	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	58.0	58.0	33.13	8.58	0.29
	38	5	26.2	20916.1	24175.0	24412.9	26392.5	26557.3	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	56.45	56.45	34.53	8.69	0.32
	38	5	26.2	20993.4	24159.1	24393.6	26377.4	26545.8	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	57.34	57.34	33.72	8.63	0.3
	38	5	26.2	21013.4	24152.1	24389.1	26370.7	26537.5	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	57.66	57.66	33.44	8.61	0.3
	38	5	26.2	20870.1	24175.2	24415.7	26387.0	26554.6	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	56.31	56.31	34.65	8.71	0.33
	38	5	26.2	20935.3	24169.3	24406.7	26383.9	26551.3	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	56.77	56.77	34.24	8.67	0.32
	38	5	19.2	21639.4	23708.1	23921.9	25461.4	25586.9	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	78.25	78.25	3.75	17.79	0.2
	38	5	19.2	20888.0	23848.7	24062.3	25678.3	25794.1	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	72.44	72.44	5.5	21.6	0.46
	38	5	19.2	21178.0	23805.3	24017.9	25622.0	25736.7	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	74.41	74.41	4.89	20.34	0.35
	38	5	19.2	22343.4	23634.1	23842.3	25314.5	25421.7	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	81.34	81.34	2.92	15.64	0.1
	38	5	20.7	21606.3	23925.0	24140.3	25919.3	26038.8	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	71.37	71.37	6.4	21.83	0.4
	38	5	20.7	22583.3	23808.9	24021.7	25744.2	25867.2	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	75.77	75.77	4.87	19.18	0.18
	38	5	19.1	20888.5	23886.5	24102.7	25701.9	25807.3	CONTAMINACAO BAIXA	CONTAMINACAO ALTA	71.48	71.48	5.84	22.17	0.51
	38	5	19.1	21063.2	23848.2	24061.9	25634.0	25745.6	CONTAMINACAO BAIXA	CONTAMINACAO ALTA	73.29	73.29	5.25	21.04	0.41

	38	5	19.1	20947.4	23869.3	24083.5	25672.6	25788.3	CONTAMINACAO BAIXA	CONTAMINACAO ALTA	72.21	72.21	5.6	21.72	0.47
	38	5	19.1	20848.0	23892.1	24109.2	25706.6	25820.4	CONTAMINACAO BAIXA	CONTAMINACAO ALTA	71.17	71.17	5.94	22.35	0.53
	38	5	19.1	22388.8	23712.2	23925.6	25403.7	25517.5	CONTAMINACAO BAIXA	CONTAMINACAO ALTA	79.26	79.26	3.48	17.12	0.13
	38	5	20.6	21144.4	24067.9	24288.7	26100.6	26229.1	CONTAMINACAO BAIXA	CONTAMINACAO ALTA	65.75	65.75	8.46	25.02	0.76
	38	5	19.0	21858.4	23889.2	24110.5	25641.9	25761.0	CONTAMINACAO BAIXA	CONTAMINACAO ALTA	73.57	73.57	5.23	20.88	0.32
	38	5	19.0	22011.5	23835.9	24056.2	25540.6	25650.9	CONTAMINACAO BAIXA	CONTAMINACAO ALTA	75.72	75.72	4.54	19.49	0.24
	38	5	19.0	21781.8	23848.3	24070.2	25571.2	25683.3	CONTAMINACAO BAIXA	CONTAMINACAO ALTA	74.97	74.97	4.78	19.97	0.28
	38	5	19.0	21666.2	23857.8	24079.5	25592.5	25707.9	CONTAMINACAO BAIXA	CONTAMINACAO ALTA	74.43	74.43	4.94	20.31	0.31
	38	5	20.4	20587.4	24236.0	24461.2	26339.9	26472.5	CONTAMINACAO BAIXA	CONTAMINACAO ALTA	57.91	57.91	11.57	28.94	1.58
	38	5	20.4	20390.1	24270.7	24497.4	26392.2	26523.0	CONTAMINACAO BAIXA	CONTAMINACAO ALTA	55.89	55.89	12.44	29.79	1.88
	38	5	18.8	20416.4	24152.5	24373.0	26136.3	26258.3	PURA	CONTAMINACAO ALTA	59.42	59.42	10.49	28.5	1.59
	38	5	18.8	20481.0	24179.1	24397.8	26208.2	26339.3	PURA	CONTAMINACAO ALTA	58.23	58.23	11.02	29.03	1.72
	38	5	18.8	20863.6	24086.3	24308.7	26012.8	26145.0	PURA	CONTAMINACAO ALTA	63.62	63.62	8.85	26.47	1.06
	38	5	18.8	21003.4	24077.7	24297.6	25989.2	26107.2	PURA	CONTAMINACAO ALTA	64.57	64.57	8.49	25.98	0.95
	38	5	19.0	20815.3	24077.0	24298.0	25999.6	26119.9	PURA	CONTAMINACAO ALTA	64.09	64.09	8.58	26.32	1.02
	38	5	20.3	19753.8	24356.9	24585.2	26540.3	26675.2	PURA	CONTAMINACAO ALTA	49.36	49.36	15.22	32.28	3.14
	38	5	20.3	20060.2	24307.1	24534.5	26462.8	26598.2	PURA	CONTAMINACAO ALTA	52.75	52.75	13.7	31.1	2.45
	41	0	25.9	19868.8	24827.1	25045.9	27247.6	27436.6	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	57.36	30.42	57.36	10.63	1.59
	41	0	25.3	18910.7	25141.9	25397.6	27731.1	27903.0	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	61.88	21.24	61.88	12.8	4.08
	41	0	25.3	18920.3	25146.3	25401.3	27739.9	27895.0	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	61.87	21.26	61.87	12.79	4.08
	41	0	25.3	18913.1	25143.2	25389.7	27718.3	27912.8	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	61.83	21.29	61.83	12.81	4.07
	41	0	25.3	18911.7	25144.4	25403.3	27738.0	27893.7	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	61.9	21.22	61.9	12.79	4.09
	41	0	25.3	18934.6	25148.1	25401.0	27740.2	27897.4	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	61.82	21.32	61.82	12.79	4.07
	41	0	25.3	18917.1	25139.7	25394.3	27725.2	27898.9	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	61.8	21.32	61.8	12.81	4.07
	41	0	25.3	18931.5	25138.6	25393.2	27722.5	27893.9	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	61.71	21.43	61.71	12.82	4.04
	41	0	25.3	18924.7	25136.2	25392.3	27721.3	27893.1	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	61.72	21.41	61.72	12.82	4.05
	41	0	25.3	18937.0	25134.0	25389.1	27713.4	27886.4	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	61.61	21.53	61.61	12.84	4.02
	41	0	25.3	18910.2	25144.2	25400.9	27735.2	27885.0	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	61.87	21.24	61.87	12.8	4.09
	41	0	25.5	18947.8	25149.8	25401.3	27748.3	27910.5	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	64.01	20.6	64.01	11.67	3.71
	41	0	25.5	18952.7	25148.5	25405.3	27758.4	27920.3	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	64.04	20.58	64.04	11.66	3.71

	41	0	25.5	18973.3	25147.0	25389.6	27719.9	27933.1	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	63.83	20.81	63.83	11.7	3.66
	41	0	25.5	19012.2	25153.9	25407.4	27779.2	27950.1	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	63.92	20.76	63.92	11.66	3.67
	41	0	25.5	18990.8	25156.1	25409.1	27779.6	27949.5	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	64.03	20.63	64.03	11.65	3.7
	41	0	25.5	18960.0	25154.4	25408.2	27767.3	27927.3	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	64.08	20.55	64.08	11.65	3.72
	41	0	25.5	18974.7	25155.1	25409.4	27774.4	27942.0	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	64.07	20.57	64.07	11.65	3.71
	41	0	25.5	19063.6	25152.1	25404.9	27773.7	27949.4	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	63.66	21.05	63.66	11.68	3.6
	41	0	25.5	18940.4	25149.4	25404.0	27749.6	27908.2	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	64.06	20.56	64.06	11.67	3.72
	41	0	25.5	19034.3	25154.0	25406.9	27778.9	27953.7	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	63.82	20.87	63.82	11.67	3.65
	41	0	17.9	20404.7	24858.5	25116.5	27182.2	27351.2	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	34.35	34.35	24.45	33.7	7.5
	41	0	18.0	21033.6	24797.0	25055.3	27094.9	27264.2	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	38.51	38.51	22.31	33.85	5.33
	41	0	17.6	18831.8	25206.4	25447.8	27764.1	27948.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	32.38	18.74	32.38	29.57	19.31
	41	0	17.8	20690.0	24875.1	25130.7	27208.0	27375.5	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	35.01	35.01	24.58	33.46	6.95
	41	0	17.8	20538.9	24906.3	25164.9	27222.4	27391.3	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	33.85	33.85	25.14	33.41	7.59
	41	0	17.8	21072.9	24910.9	25171.9	27422.4	27636.2	CONTAMINACAO MEDIA	CONTAMINACAO ALTA	34.26	34.26	25.48	33.47	6.8
	41	0	24.3	17443.1	25384.3	25561.7	28337.0	28647.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	60.31	12.69	60.31	16.07	10.93
	41	0	24.3	17507.4	25388.9	25570.7	28312.4	28604.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	60.06	13.01	60.06	16.15	10.78
	41	0	24.3	17356.9	25404.0	25580.9	28343.3	28661.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	60.72	12.16	60.72	15.88	11.24
	41	0	24.3	17484.2	25394.9	25574.4	28341.0	28638.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	60.2	12.84	60.2	16.08	10.88
	41	0	24.3	17638.0	25380.0	25548.4	28251.2	28570.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	59.45	13.8	59.45	16.41	10.35
	41	0	24.3	17516.4	25385.2	25563.7	28302.7	28618.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	60.01	13.08	60.01	16.18	10.73
	41	0	24.3	17470.3	25387.6	25565.0	28334.8	28636.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	60.21	12.82	60.21	16.09	10.87
	41	0	24.3	17406.0	25377.6	25554.2	28341.0	28651.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	60.42	12.54	60.42	16.04	11.0
	41	0	24.3	17556.2	25378.8	25555.9	28283.0	28570.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	59.79	13.35	59.79	16.28	10.58
	41	0	24.3	17425.4	25382.3	25559.7	28341.7	28651.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	60.37	12.61	60.37	16.05	10.97
	41	0	25.2	17518.9	25448.7	25628.3	28348.0	28620.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	69.95	11.24	69.95	11.17	7.65
	41	0	25.2	17479.0	25439.5	25615.0	28334.3	28613.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	70.04	11.11	70.04	11.17	7.68
	41	0	25.2	17447.5	25441.0	25617.4	28344.1	28627.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	70.17	10.96	70.17	11.12	7.74
	41	0	25.2	17405.7	25430.7	25608.1	28365.7	28660.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	70.31	10.8	70.31	11.09	7.79
	41	0	25.2	17550.3	25447.2	25626.7	28318.7	28577.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	69.79	11.41	69.79	11.23	7.57
	41	0	25.2	17421.9	25440.6	25618.1	28346.5	28634.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	70.27	10.84	70.27	11.1	7.79

	41	0	25.2	17536.0	25445.4	25623.9	28327.9	28594.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	69.85	11.34	69.85	11.21	7.6
	41	0	25.2	17580.9	25459.6	25645.5	28339.2	28619.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	69.77	11.46	69.77	11.2	7.58
	41	0	25.2	17525.5	25457.9	25633.8	28263.6	28628.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	69.92	11.26	69.92	11.18	7.63
	41	0	25.2	17498.2	25447.2	25627.2	28337.7	28629.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	70.02	11.15	70.02	11.16	7.68
	41	0	17.3	20357.1	24911.0	25162.7	27181.8	27343.1	CONTAMINACAO BAIXA	CONTAMINACAO ALTA	32.58	32.58	27.12	31.98	8.31
	41	0	17.4	20490.7	24967.7	25219.4	27370.4	27550.3	CONTAMINACAO BAIXA	CONTAMINACAO MEDIA	32.11	31.2	27.85	32.11	8.84
	41	0	17.5	20106.9	24998.6	25251.8	27381.2	27548.5	CONTAMINACAO BAIXA	CONTAMINACAO MEDIA	32.17	29.22	28.3	32.17	10.32
	41	0	17.5	20358.4	24941.6	25195.4	27282.4	27447.8	CONTAMINACAO BAIXA	CONTAMINACAO MEDIA	32.49	31.75	27.03	32.49	8.73
	41	0	19.9	20016.8	25050.2	25296.8	27447.3	27609.4	CONTAMINACAO BAIXA	CONTAMINACAO MEDIA	35.12	31.24	24.85	35.12	8.8
	41	0	19.9	20161.0	25022.2	25269.2	27409.1	27568.3	CONTAMINACAO BAIXA	CONTAMINACAO MEDIA	35.2	32.49	24.24	35.2	8.07
	41	0	19.8	18552.6	25351.1	25587.0	27936.9	28113.7	CONTAMINACAO BAIXA	CONTAMINACAO MEDIA	31.92	18.36	30.5	31.92	19.22
	41	0	19.8	18409.3	25359.8	25595.6	27944.8	28114.9	CONTAMINACAO BAIXA	CONTAMINACAO MEDIA	31.6	17.5	30.8	31.6	20.1
	41	0	17.8	18466.6	24865.6	25117.4	26976.8	27113.1	PURA	CONTAMINACAO MEDIA	33.36	24.85	28.2	33.36	13.58
	41	0	18.0	18806.4	24778.3	25038.3	26798.1	26930.5	PURA	CONTAMINACAO MEDIA	34.49	30.09	25.12	34.49	10.3
	41	0	18.0	18603.4	24787.5	25047.1	26817.0	26956.7	PURA	CONTAMINACAO MEDIA	34.47	28.38	25.9	34.47	11.25
	41	0	18.3	18796.0	24698.2	24936.4	26865.5	27048.8	PURA	CONTAMINACAO MEDIA	35.25	30.84	24.11	35.25	9.8
	41	0	18.6	18374.0	25173.8	25414.5	27613.9	27757.1	PURA	CONTAMINACAO MEDIA	32.03	18.44	30.08	32.03	19.45
	41	0	19.7	18064.4	25430.2	25663.4	28072.5	28253.2	PURA	CONTAMINACAO BAIXA	31.68	14.83	31.68	30.16	23.33
	41	0	19.7	18287.4	25372.1	25608.4	27973.4	28152.2	PURA	CONTAMINACAO MEDIA	31.18	16.57	31.03	31.18	21.22
	39	0	18.6	18019.4	25796.2	25963.9	28595.8	28821.4	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	32.67	11.06	32.67	24.58	31.68
	39	0	18.6	19164.4	25625.8	25830.7	28357.3	28576.8	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	31.67	17.12	31.67	28.52	22.7
	39	0	18.6	18449.5	25749.3	25942.6	28536.1	28751.0	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	32.41	13.12	32.41	25.99	28.48
	39	0	18.6	18743.0	25717.1	25915.3	28493.7	28717.5	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	32.19	14.62	32.19	26.93	26.26
	39	0	18.6	18531.4	25725.4	25921.5	28490.4	28703.6	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	32.32	13.6	32.32	26.37	27.71
	39	0	19.1	17281.4	26207.0	26384.7	29150.9	29405.9	CONTAMINACAO ALTA	PURA	37.55	8.21	32.86	21.38	37.55
	39	0	19.1	17548.1	26107.9	26274.6	29101.2	29374.4	CONTAMINACAO ALTA	PURA	35.32	9.3	32.81	22.57	35.32
	39	0	25.9	15845.8	25860.5	25928.5	28662.7	29036.8	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	82.44	3.56	82.44	5.47	8.52
	39	0	25.7	17399.1	25968.4	26120.0	29208.2	29591.9	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	76.96	8.39	76.96	7.32	7.33
	39	0	25.7	17354.5	25960.7	26103.8	29188.9	29580.4	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	77.09	8.22	77.09	7.3	7.39
	39	0	25.7	17336.5	25964.0	26103.4	29187.7	29579.4	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	77.14	8.15	77.14	7.29	7.42

	39	0	25.7	17371.9	25963.1	26111.8	29198.2	29584.0	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	77.04	8.29	77.04	7.31	7.36
	39	0	25.7	17405.5	25969.4	26123.6	29208.3	29589.2	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	76.95	8.4	76.95	7.33	7.32
	39	0	25.7	17391.1	25965.4	26116.0	29203.0	29585.9	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	76.99	8.35	76.99	7.32	7.34
	39	0	25.7	17380.8	25966.2	26115.4	29201.8	29586.1	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	77.02	8.32	77.02	7.31	7.35
	39	0	25.7	17363.3	25964.3	26110.5	29196.2	29583.7	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	77.07	8.25	77.07	7.3	7.38
	39	0	25.7	17414.3	25966.3	26120.0	29205.4	29586.2	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	76.92	8.44	76.92	7.34	7.31
	39	0	25.7	17342.8	25962.3	26106.3	29190.7	29577.8	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	77.12	8.18	77.12	7.29	7.41
	39	0	26.0	17324.9	25956.5	26093.5	29174.4	29566.9	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	80.03	7.5	80.03	6.16	6.31
	39	0	26.0	17355.9	25906.9	26027.0	29062.6	29462.8	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	79.94	7.56	79.94	6.24	6.26
	39	0	26.0	17283.7	25952.5	26082.7	29154.7	29553.7	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	80.15	7.35	80.15	6.14	6.36
	39	0	26.0	17408.6	25889.7	26003.5	29025.1	29428.2	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	79.79	7.72	79.79	6.3	6.19
	39	0	26.0	17326.8	25919.2	26047.3	29099.1	29503.4	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	80.02	7.48	80.02	6.2	6.29
	39	0	26.0	17302.2	25956.1	26092.6	29169.6	29563.4	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	80.1	7.42	80.1	6.15	6.34
	39	0	26.0	17291.1	25953.8	26086.3	29160.5	29557.8	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	80.13	7.38	80.13	6.14	6.35
	39	0	26.0	17303.8	25936.5	26061.8	29121.0	29522.1	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	80.09	7.4	80.09	6.17	6.33
	39	0	26.0	17312.4	25958.0	26094.3	29174.1	29567.5	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	80.07	7.46	80.07	6.15	6.33
	39	0	26.0	17288.9	25943.9	26072.4	29138.5	29538.4	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	80.13	7.36	80.13	6.15	6.35
	39	0	18.6	18418.9	25704.6	25874.4	28358.7	28570.2	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	32.29	13.26	32.29	26.41	28.04
	39	0	18.6	18787.8	25528.0	25713.8	28187.1	28401.0	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	31.6	16.04	31.6	28.52	23.83
	39	0	18.7	18585.9	25610.7	25791.0	28292.2	28499.2	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	31.89	14.62	31.89	27.63	25.87
	39	0	18.7	18603.1	25619.0	25802.8	28218.1	28413.6	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	31.83	14.75	31.83	27.73	25.68
	39	0	19.2	17732.7	26023.8	26206.3	29105.2	29397.8	CONTAMINACAO ALTA	PURA	33.3	10.31	32.78	23.61	33.3
	39	0	18.8	17120.8	25844.0	25927.1	28451.4	28687.5	CONTAMINACAO MEDIA	PURA	37.15	7.61	32.87	22.37	37.15
	39	0	18.8	18341.6	25611.4	25757.7	28196.7	28406.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	31.91	13.73	31.91	27.45	26.92
	39	0	18.8	17570.1	25777.6	25889.1	28384.0	28605.6	CONTAMINACAO MEDIA	PURA	33.71	9.52	32.65	24.13	33.71
	39	0	18.8	17128.9	25860.1	25940.1	28481.8	28720.1	CONTAMINACAO MEDIA	PURA	37.26	7.6	32.87	22.28	37.26
	39	0	18.8	18411.4	25598.1	25742.2	28175.5	28389.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	31.81	14.16	31.81	27.74	26.29
	39	0	19.1	17437.5	26004.6	26152.9	29092.2	29400.5	CONTAMINACAO MEDIA	PURA	35.28	9.04	32.92	22.77	35.28
	39	0	19.1	17606.3	25972.3	26128.4	29069.8	29383.0	CONTAMINACAO MEDIA	PURA	34.02	9.75	32.84	23.39	34.02
	39	0	22.5	17041.6	25409.1	25447.2	28013.6	28377.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	44.86	11.52	44.86	23.45	20.17

	39	0	22.5	16986.9	25413.4	25451.1	28015.4	28380.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	45.02	11.23	45.02	23.3	20.45
	39	0	22.5	17070.6	25411.0	25447.7	28014.4	28380.2	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	44.78	11.66	44.78	23.5	20.06
	39	0	22.5	16963.6	25409.6	25443.0	28008.1	28373.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	45.06	11.16	45.06	23.3	20.48
	39	0	22.5	16929.8	25420.1	25455.5	28025.5	28391.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	45.2	10.91	45.2	23.13	20.76
	39	0	22.5	16907.8	25416.5	25448.9	28012.7	28379.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	45.24	10.85	45.24	23.13	20.79
	39	0	22.5	16928.0	25412.0	25448.2	28012.6	28380.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	45.17	10.96	45.17	23.19	20.68
	39	0	22.5	17169.5	25405.8	25443.0	27973.3	28367.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	44.46	12.23	44.46	23.78	19.52
	39	0	22.5	17113.6	25407.1	25446.2	28008.4	28377.9	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	44.65	11.89	44.65	23.61	19.84
	39	0	22.5	16947.0	25409.4	25445.7	28010.7	28375.3	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	45.11	11.07	45.11	23.25	20.57
	39	0	23.4	16755.8	25381.6	25408.3	27924.0	28277.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	53.18	10.4	53.18	19.58	16.85
	39	0	23.4	16837.8	25404.4	25437.6	27987.3	28352.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	53.07	10.55	53.07	19.5	16.88
	39	0	23.4	16627.6	25335.8	25351.2	27849.2	28193.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	53.35	10.19	53.35	19.73	16.74
	39	0	23.4	16732.7	25368.3	25389.3	27906.2	28256.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	53.18	10.4	53.18	19.65	16.77
	39	0	23.4	16665.7	25347.4	25363.8	27868.3	28217.9	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	53.28	10.27	53.28	19.7	16.75
	39	0	23.4	16814.2	25389.1	25417.2	27970.8	28333.4	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	53.07	10.55	53.07	19.57	16.81
	39	0	23.4	16694.9	25357.7	25377.3	27890.2	28239.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	53.25	10.31	53.25	19.66	16.78
	39	0	23.4	16860.2	25407.4	25436.6	27991.0	28354.8	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	53.0	10.64	53.0	19.53	16.82
	39	0	23.4	16764.8	25376.7	25400.9	27919.7	28271.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	53.12	10.48	53.12	19.64	16.76
	39	0	23.4	16766.4	25394.8	25424.2	27926.0	28297.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	53.2	10.36	53.2	19.51	16.93
	39	0	15.4	16572.4	26106.8	26200.6	28837.3	29085.9	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	45.34	3.52	45.34	10.15	41.0
	39	0	15.4	17614.3	25967.2	26103.8	28652.5	28866.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	46.39	6.08	46.39	12.52	35.02
	39	0	15.7	17610.4	25897.0	26025.4	28574.9	28809.7	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	44.24	6.5	44.24	13.92	35.35
	39	0	15.9	17243.4	25995.6	26110.0	28709.1	28945.6	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	42.51	5.5	42.51	13.48	38.51
	39	0	19.1	16288.9	26366.2	26523.5	29461.5	29727.6	CONTAMINACAO MEDIA	PURA	43.02	5.29	33.37	18.32	43.02
	39	0	19.1	16467.9	26319.0	26474.8	29413.9	29690.4	CONTAMINACAO MEDIA	PURA	42.0	5.77	33.3	18.93	42.0
	39	0	19.1	16794.0	26079.1	26204.6	28963.8	29262.5	CONTAMINACAO BAIXA	PURA	40.13	6.44	33.01	20.42	40.13
	39	0	19.1	16931.1	26051.0	26176.6	28906.0	29200.2	CONTAMINACAO BAIXA	PURA	39.22	6.88	32.96	20.95	39.22
	39	0	19.5	16461.1	26171.8	26270.6	29054.5	29402.1	CONTAMINACAO BAIXA	PURA	41.26	5.64	33.47	19.63	41.26
	39	0	19.5	16249.6	26236.4	26332.3	29141.0	29495.8	CONTAMINACAO BAIXA	PURA	42.54	5.07	33.57	18.82	42.54
	39	0	15.4	16285.4	26329.3	26414.4	28995.7	29216.8	PURA	CONTAMINACAO BAIXA	44.71	2.93	44.71	9.17	43.18

	39	0	15.4	16470.1	26301.4	26392.4	28975.0	29317.3	PURA	CONTAMINACAO BAIXA	45.0	3.29	45.0	9.56	42.15
	39	0	15.5	16439.5	26291.1	26380.4	28985.7	29187.5	PURA	CONTAMINACAO BAIXA	44.18	3.27	44.18	9.8	42.76
	39	0	15.5	16753.3	26185.2	26272.1	28767.1	28966.1	PURA	CONTAMINACAO BAIXA	44.5	3.86	44.5	10.61	41.03
	39	0	15.5	16485.6	26266.2	26352.2	28937.2	29130.3	PURA	CONTAMINACAO BAIXA	44.22	3.34	44.22	9.93	42.52
	39	0	26.2	15810.8	25663.1	25674.9	28076.0	28556.7	PURA	CONTAMINACAO BAIXA	84.88	3.52	84.88	4.94	6.66
	34	15	19.3	18669.9	25898.8	26085.2	28963.2	29262.9	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	30.94	15.31	30.94	26.23	27.52
	34	15	19.3	18751.2	25889.5	26070.7	28974.8	29279.1	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	30.88	15.78	30.88	26.48	26.85
	34	15	19.3	18669.4	25911.0	26098.5	29009.9	29314.0	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	30.95	15.36	30.95	26.18	27.5
	34	15	19.3	18951.7	25833.1	26021.3	28892.1	29197.4	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	30.68	16.85	30.68	27.2	25.27
	34	15	19.3	18694.8	25901.5	26089.7	28992.9	29294.9	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	30.93	15.49	30.93	26.28	27.31
	34	15	21.5	18180.8	25922.0	26052.3	28767.2	29034.5	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	36.57	14.67	36.57	24.5	24.27
	34	15	19.1	16663.5	26163.3	26299.1	29315.2	29690.0	CONTAMINACAO ALTA	PURA	41.85	6.61	31.94	19.6	41.85
	34	15	19.1	17784.8	26014.6	26158.3	29032.1	29346.1	CONTAMINACAO ALTA	PURA	34.66	10.72	31.42	23.2	34.66
	34	15	19.1	17122.6	26069.9	26203.1	29172.4	29534.9	CONTAMINACAO ALTA	PURA	38.95	8.14	31.73	21.18	38.95
	34	15	19.1	17872.4	25992.3	26136.1	28999.1	29310.6	CONTAMINACAO ALTA	PURA	33.98	11.11	31.38	23.53	33.98
	34	15	21.5	16360.9	26214.3	26263.6	28945.6	29238.3	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	38.72	6.36	38.72	18.82	36.11
	34	15	21.5	16421.1	26190.2	26235.9	28859.9	29129.8	CONTAMINACAO ALTA	CONTAMINACAO BAIXA	38.62	6.52	38.62	19.06	35.81
	34	15	19.2	16633.4	26179.7	26298.3	29232.8	29564.9	CONTAMINACAO MEDIA	PURA	42.29	6.39	31.87	19.45	42.29
	34	15	19.2	16598.5	26199.4	26317.5	29262.2	29596.8	CONTAMINACAO MEDIA	PURA	42.5	6.3	31.89	19.31	42.5
	34	15	19.2	16858.5	26123.1	26238.1	29104.8	29432.2	CONTAMINACAO MEDIA	PURA	41.0	7.04	31.74	20.23	41.0
	34	15	19.2	16692.7	26161.8	26275.8	29192.1	29526.3	CONTAMINACAO MEDIA	PURA	41.96	6.55	31.83	19.66	41.96
	34	15	19.2	16575.5	26199.3	26319.6	29264.2	29595.7	CONTAMINACAO MEDIA	PURA	42.64	6.23	31.89	19.24	42.64
	34	15	21.5	16044.4	26041.3	26090.7	28631.1	28880.0	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	38.99	5.4	38.99	18.65	36.95
	34	15	21.5	16199.9	25997.7	26046.9	28613.4	28852.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	38.87	5.94	38.87	19.31	35.88
	34	15	19.2	17004.1	25898.3	25951.0	28620.9	28906.6	CONTAMINACAO MEDIA	PURA	38.98	7.61	31.62	21.79	38.98
	34	15	19.2	17142.5	25864.5	25922.1	28577.0	28866.3	CONTAMINACAO MEDIA	PURA	37.81	8.2	31.57	22.42	37.81
	34	15	19.2	17404.4	25734.6	25780.6	28405.3	28727.2	CONTAMINACAO MEDIA	PURA	34.73	9.68	31.44	24.15	34.73
	34	15	19.2	17697.7	25859.5	25945.6	28662.9	28944.6	CONTAMINACAO MEDIA	PURA	34.18	10.49	31.34	23.99	34.18
	34	15	19.2	17612.6	25881.3	25965.3	28694.8	28976.6	CONTAMINACAO MEDIA	PURA	34.92	10.07	31.39	23.61	34.92
	34	15	21.4	17079.2	25733.8	25770.9	28239.2	28457.5	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	37.21	10.06	37.21	23.7	29.03

	34	15	21.4	16904.2	25790.7	25824.2	28344.2	28610.1	CONTAMINACAO MEDIA	CONTAMINACAO BAIXA	37.54	9.02	37.54	22.74	30.7
	34	15	19.1	15882.8	26147.1	26203.8	28978.1	29309.3	CONTAMINACAO BAIXA	PURA	47.15	4.01	31.8	17.04	47.15
	34	15	19.1	16935.6	26204.2	26248.7	28952.4	29262.3	CONTAMINACAO BAIXA	PURA	41.56	6.98	31.5	19.96	41.56
	34	15	19.1	15928.7	26129.8	26176.0	28804.7	29119.4	CONTAMINACAO BAIXA	PURA	47.1	4.04	31.67	17.19	47.1
	34	15	19.1	16829.9	26249.9	26292.0	29022.2	29331.3	CONTAMINACAO BAIXA	PURA	42.34	6.64	31.52	19.5	42.34
	34	15	21.3	15883.9	26031.8	26072.2	28754.8	29073.7	CONTAMINACAO BAIXA	PURA	38.53	4.92	38.15	18.4	38.53
	34	15	19.1	16981.7	25822.1	25841.2	28383.0	28649.9	CONTAMINACAO BAIXA	PURA	38.39	7.7	31.6	22.31	38.39
	34	15	19.1	16219.5	25910.4	25901.8	28400.2	28673.8	CONTAMINACAO BAIXA	PURA	43.66	5.05	31.84	19.46	43.66
	34	15	19.1	16901.5	25846.5	25862.5	28412.5	28679.3	CONTAMINACAO BAIXA	PURA	39.15	7.34	31.62	21.89	39.15
	34	15	19.1	17383.0	25700.9	25737.1	28238.0	28495.6	CONTAMINACAO BAIXA	PURA	34.43	9.76	31.36	24.44	34.43
	34	15	19.1	16102.0	25915.0	25906.2	28431.7	28792.4	CONTAMINACAO BAIXA	PURA	44.39	4.71	31.9	19.0	44.39

Registro Tecnico e Metadados

```
--- DADOS DA EXECUÇÃO ---
Data/Hora: 21/06/2026 19:20
Target: Classe
Features: TeorRotulo, TeorDensimetro, NivelAcucar, Temperatura, CPJ_10k, CPJ_4k7, CPJ_3k19, CPJ_1k, CPJ_824R
Dados de aprendizagem: 1841 | Dados externos de teste: 303
Cross-Validation: 12 Folds
--- AVALIAÇÃO DE COMPLEXIDADE DOS DADOS ---
Espectro Frequencial (Classes): CONTAMINACAO BAIXA: 29.0% | CONTAMINACAO ALTA: 28.7% | CONTAMINACAO MEDIA: 28.4% | PURA: 14.0%
■ Frequência da classe minoritária: 257 amostras.
Diagnóstico de Multicolinearidade
  - Condition Number (CN): 228697.35 - CN > 30 são consideradas colineares, sinalizando instabilidade matricial.
  - Mean Variance Inflation Factor (VIF): 430878.98 - quanto maior o valor (acima de 5), maior a redundância de dados.
Separabilidade Geométrica das Classes (Silhouette Score): -0.062
--- AVALIAÇÃO DE MODELOS ---
MÉTRICAS DE DESEMPENHO:
  - F1-Macro: média harmônica entre precisão e recall (calculada para cada classe e promediada); excelente para medir eficiência em bases desbalanceadas.
  - Acc (Accuracy): proporção de amostras classificadas corretamente; mais intuitiva, porém sensível a classes majoritárias.
  - B-Acc (Balanced Accuracy): média do recall de cada classe individualmente; ajusta a percepção de acertos compensando distorções de frequência de classe.
DIAGNÓSTICO ESPACIAL E DE CLASSES:
  - Matriz de Confusão: tabulação cruzada das predições corretas e incorretas detalhadas pela realidade amostral.
  - Frequência e Silhouette Score: análise da dominância das classes e da sobreposição vetorial geométrica das predições.
> Modelo: Regressão Logística
  Melhores Hiperparâmetros: {'C': 100.0, 'class_weight': None, 'solver': 'lbfgs'}
  Tempo de execução: 15.66s
  Validação Cruzada: F1-Macro = 0.6895 | Acc = 0.6975 | B-Acc = 0.6876
  Validação Cega: F1-Macro = 0.6636 | Acc = 0.6667 | B-Acc = 0.6830
  ■ White-Box (Logística - Pesos): Z = (24.013 * TeorRotulo) + (69.363 * TeorDensimetro) + (75.585 * NivelAcucar) + (0.468 * Temperatura) + (2.873 * CPJ_10k) + (-27.380 * CPJ_4k7) + (17.4
44 * CPJ_3k19) + (13.999 * CPJ_1k) + (-7.573 * CPJ_824R)
  Distribuição de Acertos (Matriz de Confusão do Teste Cego):
    Eixos (Classes): ['CONTAMINACAO ALTA', 'CONTAMINACAO BAIXA', 'CONTAMINACAO MEDIA', 'PURA']
    Real CONTAMINACAO ALTA: [92, 2, 18, 0]
    Real CONTAMINACAO BAIXA: [1, 41, 8, 2]
    Real CONTAMINACAO MEDIA: [30, 27, 51, 4]
    Real PURA: [0, 9, 0, 18]
> Modelo: SVM
  Melhores Hiperparâmetros: {'C': 100, 'class_weight': None, 'gamma': 1.0, 'kernel': 'rbf'}
  Tempo de execução: 289.98s
  Validação Cruzada: F1-Macro = 0.9952 | Acc = 0.9957 | B-Acc = 0.9943
  Validação Cega: F1-Macro = 0.7854 | Acc = 0.7525 | B-Acc = 0.8185
  Distribuição de Acertos (Matriz de Confusão do Teste Cego):
    Eixos (Classes): ['CONTAMINACAO ALTA', 'CONTAMINACAO BAIXA', 'CONTAMINACAO MEDIA', 'PURA']
    Real CONTAMINACAO ALTA: [92, 0, 20, 0]
    Real CONTAMINACAO BAIXA: [0, 52, 0, 0]
    Real CONTAMINACAO MEDIA: [15, 38, 59, 0]
    Real PURA: [0, 2, 0, 25]
> Modelo: Random Forest
  Melhores Hiperparâmetros: {'class_weight': None, 'max_depth': None, 'min_samples_split': 2, 'n_estimators': 50}
  Tempo de execução: 176.22s
  Validação Cruzada: F1-Macro = 0.9875 | Acc = 0.9875 | B-Acc = 0.9866
  Validação Cega: F1-Macro = 0.7998 | Acc = 0.7690 | B-Acc = 0.8297
  Distribuição de Acertos (Matriz de Confusão do Teste Cego):
    Eixos (Classes): ['CONTAMINACAO ALTA', 'CONTAMINACAO BAIXA', 'CONTAMINACAO MEDIA', 'PURA']
    Real CONTAMINACAO ALTA: [89, 0, 23, 0]
    Real CONTAMINACAO BAIXA: [0, 52, 0, 0]
    Real CONTAMINACAO MEDIA: [0, 45, 67, 0]
    Real PURA: [0, 2, 0, 25]
> Modelo: Gradient Boosting
```

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Melhores Hiperparâmetros: {'learning_rate': 0.1, 'max_depth': 10, 'n_estimators': 200, 'subsample': 0.7}
Tempo de execução: 1394.43s
Validação Cruzada: F1-Macro = 0.9895 | Acc = 0.9902 | B-Acc = 0.9886
Validação Cega: F1-Macro = 0.8161 | Acc = 0.7921 | B-Acc = 0.8383
Distribuição de Acertos (Matriz de Confusão do Teste Cego):
  Eixos (Classes): ['CONTAMINACAO ALTA', 'CONTAMINACAO BAIXA', 'CONTAMINACAO MEDIA', 'PURA']
  Real CONTAMINACAO ALTA: [92, 0, 20, 0]
  Real CONTAMINACAO BAIXA: [0, 52, 0, 0]
  Real CONTAMINACAO MEDIA: [9, 31, 72, 0]
  Real PURA: [0, 3, 0, 24]
> Modelo: ANN
Melhores Hiperparâmetros: {'activation': 'tanh', 'alpha': 0.1, 'hidden_layer_sizes': (100, 100), 'solver': 'lbfgs'}
Tempo de execução: 478.76s
Validação Cruzada: F1-Macro = 0.9966 | Acc = 0.9973 | B-Acc = 0.9957
Validação Cega: F1-Macro = 0.8179 | Acc = 0.7954 | B-Acc = 0.8476
Distribuição de Acertos (Matriz de Confusão do Teste Cego):
  Eixos (Classes): ['CONTAMINACAO ALTA', 'CONTAMINACAO BAIXA', 'CONTAMINACAO MEDIA', 'PURA']
  Real CONTAMINACAO ALTA: [92, 0, 20, 0]
  Real CONTAMINACAO BAIXA: [0, 52, 0, 0]
  Real CONTAMINACAO MEDIA: [4, 35, 72, 1]
  Real PURA: [0, 2, 0, 25]
> Modelo: Naive Bayes
Melhores Hiperparâmetros: Padrão
Tempo de execução: 0.12s
Validação Cruzada: F1-Macro = 0.2776 | Acc = 0.2992 | B-Acc = 0.3017
Validação Cega: F1-Macro = 0.2261 | Acc = 0.2541 | B-Acc = 0.2674
Distribuição de Acertos (Matriz de Confusão do Teste Cego):
  Eixos (Classes): ['CONTAMINACAO ALTA', 'CONTAMINACAO BAIXA', 'CONTAMINACAO MEDIA', 'PURA']
  Real CONTAMINACAO ALTA: [46, 59, 0, 7]
  Real CONTAMINACAO BAIXA: [13, 18, 7, 14]
  Real CONTAMINACAO MEDIA: [38, 51, 6, 17]
  Real PURA: [7, 7, 6, 7]
--- INFERÊNCIA ESTATÍSTICA (FRIEDMAN TEST) ---
Metodologia (Blocos): Stratified Shuffle Split (30 rodadas)
Grupos Avaliados: Regressão Logística, SVM, Random Forest, Gradient Boosting, ANN, Naive Bayes
Estatística de Teste: 140.531 | Valor-p: 1.3797e-28
Hipótese Nula (H0): Os algoritmos possuem distribuições de desempenho (F1-Macro) estatisticamente equivalentes.
Decisão: Rejeita-se H0. Há evidências estatísticas de superioridade de desempenho de um ou mais modelos.
Tempo Computacional: 42m 59.1s
-----
RAIO-X COMBINATÓRIO
--- Regressão Logística ---
* Tentativa {'C': 0.001, 'class_weight': None, 'solver': 'lbfgs'} -> F1-Macro = 0.2230
* Tentativa {'C': 0.001, 'class_weight': None, 'solver': 'liblinear'} -> F1-Macro = 0.2267
* Tentativa {'C': 0.001, 'class_weight': None, 'solver': 'saga'} -> F1-Macro = 0.2230
* Tentativa {'C': 0.001, 'class_weight': 'balanced', 'solver': 'lbfgs'} -> F1-Macro = 0.2302
* Tentativa {'C': 0.001, 'class_weight': 'balanced', 'solver': 'liblinear'} -> F1-Macro = 0.2771
* Tentativa {'C': 0.001, 'class_weight': 'balanced', 'solver': 'saga'} -> F1-Macro = 0.2298
* Tentativa {'C': 0.01, 'class_weight': None, 'solver': 'lbfgs'} -> F1-Macro = 0.3382
* Tentativa {'C': 0.01, 'class_weight': None, 'solver': 'liblinear'} -> F1-Macro = 0.3352
* Tentativa {'C': 0.01, 'class_weight': None, 'solver': 'saga'} -> F1-Macro = 0.3382
* Tentativa {'C': 0.01, 'class_weight': 'balanced', 'solver': 'lbfgs'} -> F1-Macro = 0.3632
* Tentativa {'C': 0.01, 'class_weight': 'balanced', 'solver': 'liblinear'} -> F1-Macro = 0.4527
* Tentativa {'C': 0.01, 'class_weight': 'balanced', 'solver': 'saga'} -> F1-Macro = 0.3632
* Tentativa {'C': 0.1, 'class_weight': None, 'solver': 'lbfgs'} -> F1-Macro = 0.4619
* Tentativa {'C': 0.1, 'class_weight': None, 'solver': 'liblinear'} -> F1-Macro = 0.4128
* Tentativa {'C': 0.1, 'class_weight': None, 'solver': 'saga'} -> F1-Macro = 0.4625
* Tentativa {'C': 0.1, 'class_weight': 'balanced', 'solver': 'lbfgs'} -> F1-Macro = 0.4537
* Tentativa {'C': 0.1, 'class_weight': 'balanced', 'solver': 'liblinear'} -> F1-Macro = 0.4562
* Tentativa {'C': 0.1, 'class_weight': 'balanced', 'solver': 'saga'} -> F1-Macro = 0.4551
```


[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

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--- ANN ---
* Tentativa {'activation': 'tanh', 'alpha': 0.0001, 'hidden_layer_sizes': (100,), 'solver': 'lbfgs'} -> F1-Macro = 0.9784
* Tentativa {'activation': 'tanh', 'alpha': 0.0001, 'hidden_layer_sizes': (50, 50), 'solver': 'lbfgs'} -> F1-Macro = 0.9771
* Tentativa {'activation': 'tanh', 'alpha': 0.0001, 'hidden_layer_sizes': (100, 100), 'solver': 'lbfgs'} -> F1-Macro = 0.9788
* Tentativa {'activation': 'tanh', 'alpha': 0.1, 'hidden_layer_sizes': (100,), 'solver': 'lbfgs'} -> F1-Macro = 0.9930
* Tentativa {'activation': 'tanh', 'alpha': 0.1, 'hidden_layer_sizes': (50, 50), 'solver': 'lbfgs'} -> F1-Macro = 0.9947
* Tentativa {'activation': 'tanh', 'alpha': 0.1, 'hidden_layer_sizes': (100, 100), 'solver': 'lbfgs'} -> F1-Macro = 0.9959
--- Naive Bayes ---
* Execução Única: Padrão -> Concluída.
```